COMP0222 Simultaneous Localisation and Mapping

Lecture 08: Factor Graph SLAM

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Graphical Models Redux

- We have seen we can set up estimation problems in graphs
- The graphs provide an exact description of the probability distribution
- Maximum likelihood estimation is a way to extract estimates without integration
- So…. How do we use it in SLAM?

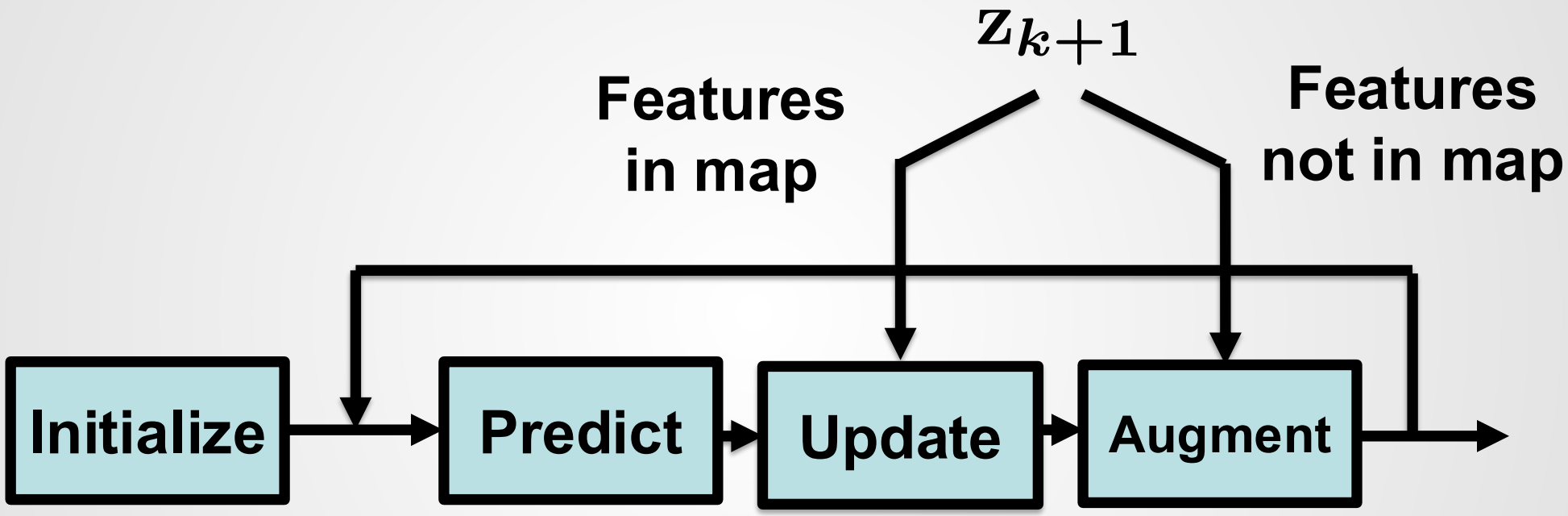
Structure

- Formulating SLAM as a Graph Problem
- SLAM Examples
- Computational and Storage Costs

Formulating SLAM as a Graph Problem

- **Formulating SLAM as a Graph Problem**
- ***SLAM Examples***
- ***Computational and Storage Costs***

SLAM Structure



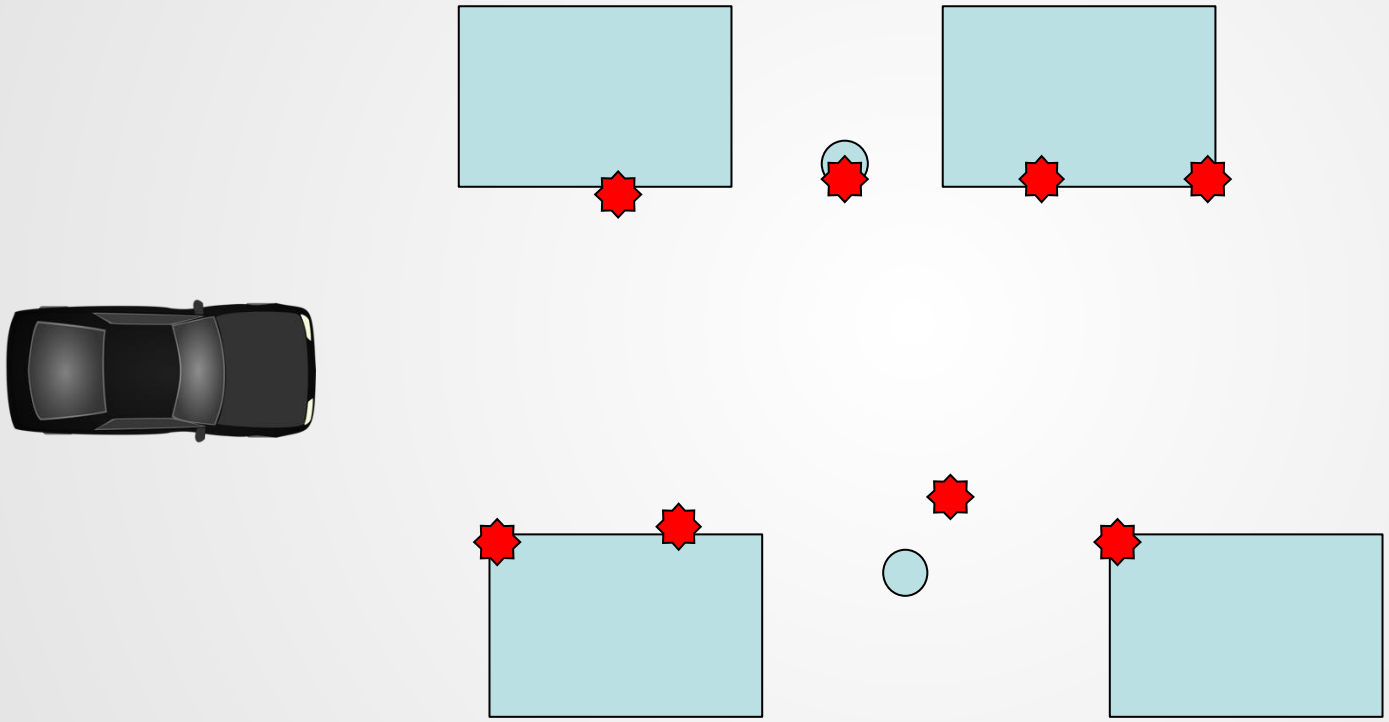
How Do We Use Graphical Models for SLAM?

- Factor Graph based SLAM broadly follows the structure of Kalman filter-based SLAM
- However, the implementation is quite a bit easier – the graph internally manages all the stuff like the cross correlations we had to explicitly work out before
- We'll follow the same pattern we used with KF-SLAM

Factor Graph SLAM: The First Few Timesteps

Timestep	Action
0	Initialize
1	Predict and initalize landmarks 1-4
2	Predict and update landmarks 4, initialize landmarks 5-8

Step 0: Initialize



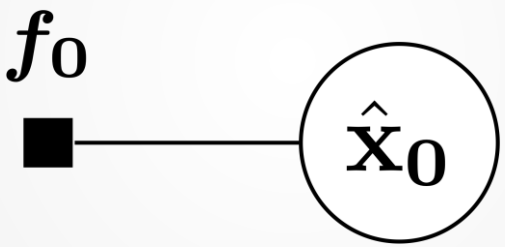
Step 0: Initialization

- The platform starts with an empty map
- Therefore, the only thing we have in the graph is a vertex which corresponds to the state of the platform at time step 0:

$$\hat{\mathbf{x}}_0$$

Step 0: Initialization Graph Structure

- **We can introduce the prior as a factor or a unary edge**



Step 0: Initialization Factor

- **Assuming a Gaussian prior,**

$$f\left(\hat{\mathbf{x}}_0\right) = \mathcal{G}\left(\hat{\mathbf{x}}_0; \bar{\mathbf{x}}_0, \mathbf{P}_0\right)$$

- **The factor is**

$$f\left(\mathbf{x}_0\right) \propto \exp\left\{-\frac{1}{2}\left(\mathbf{x}_0 \boxminus \bar{\mathbf{x}}_0\right)^\top \mathbf{P}_0^{-1}\left(\mathbf{x}_0 \boxminus \bar{\mathbf{x}}_0\right)\right\}$$

Step 0: Initializing the State

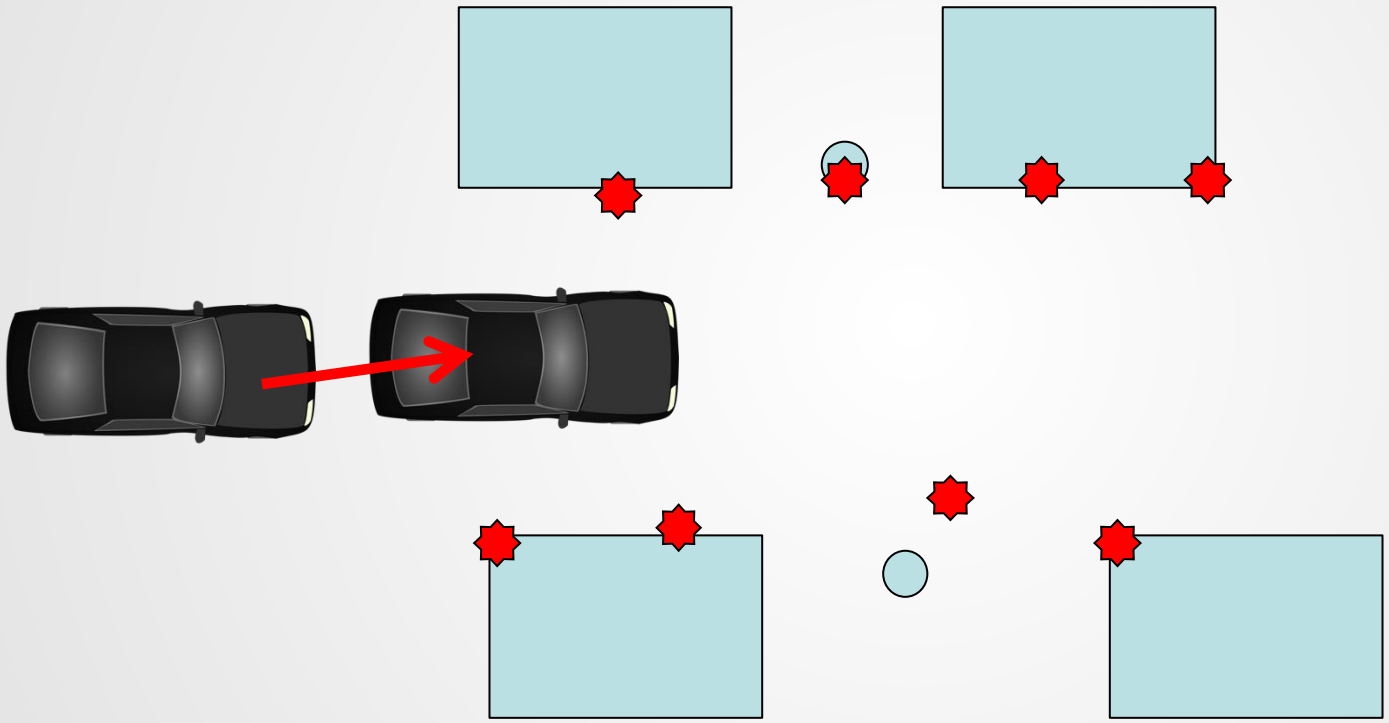
- We have to assign a value to the initial state
- We do this by setting it equal to the mean from the initial condition

$$\hat{\mathbf{x}}_0 = \bar{\mathbf{x}}_0$$

Factor Graph SLAM: The First Few Timesteps

Timestep	Action
~~0~~	~~Initialize~~
1	Predict and initalize landmarks 1-4
2	Predict and update landmarks 4, initialize landmarks 5-8

Step 1: Predict

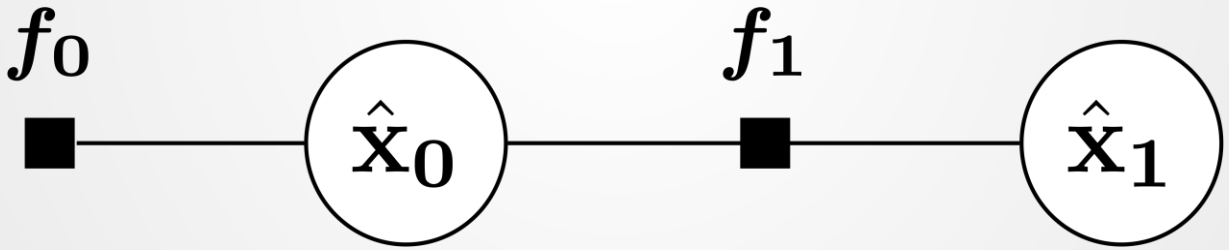


Step 1: Prediction

- We now want to estimate the platform position at time step 1
- Recall that with the graph want to *jointly* estimate the state of the first two steps
- Therefore, the graph needs to contain two platform vertices for the two timesteps

Step 1: Prediction Graph Structure

- We follow the same procedure as the tracking example - create a new vertex for the state at time step 1, and add a (binary) prediction edge



Step 1: Prediction Factor

- Recall the process model is

$$\mathbf{x}_{k+1} = \mathbf{f}\left[\mathbf{x}_k, \mathbf{u}_{k+1}, \mathbf{v}_{k+1}\right]$$

- The factor for it is

$$f\left(\mathbf{x}_{k+1} \middle| \mathbf{x}_k; \mathbf{u}_{k+1}\right) = f\left(\mathbf{v}_{k+1} = \mathbf{e}\left[\mathbf{x}_k, \mathbf{u}_{k+1}, \mathbf{x}_{k+1}\right]\right)$$

Step 1: Prediction Factor

- **Therefore, the factor has the form**

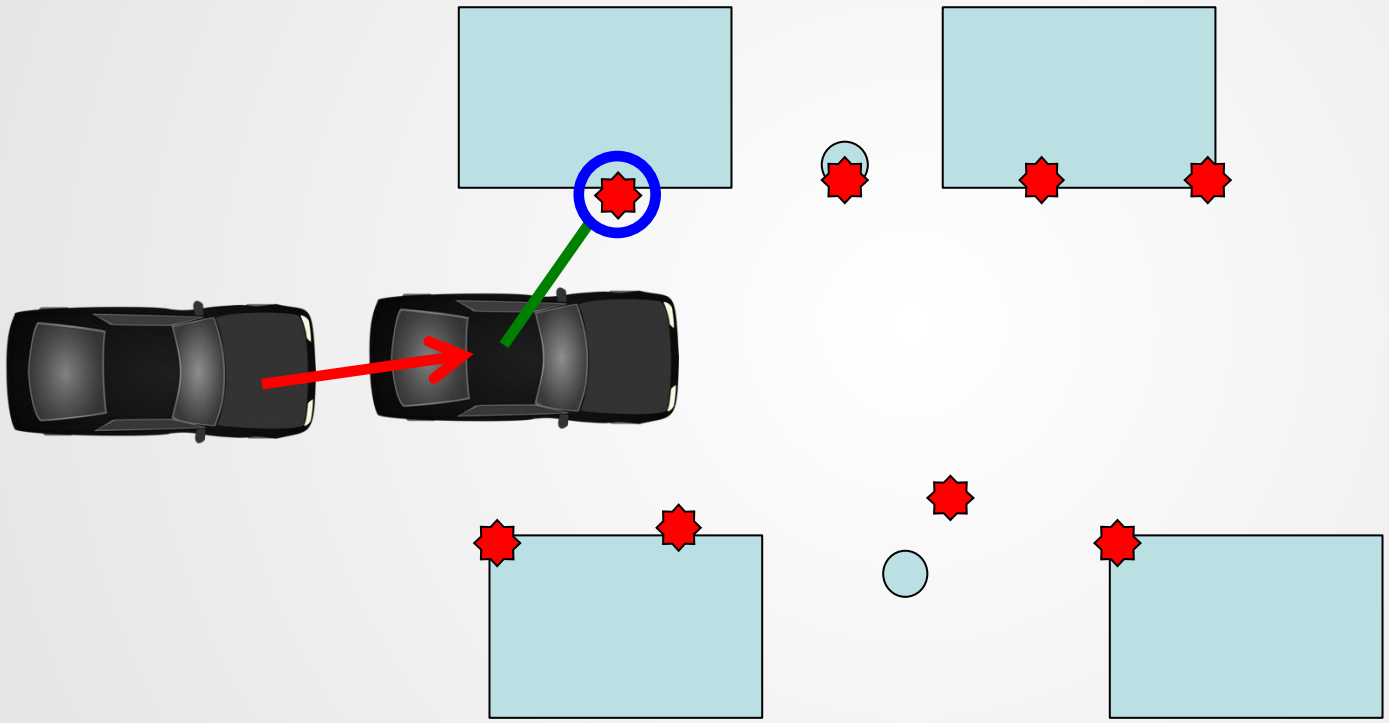
$$f\left(\hat{\mathbf{x}}_{k+1} \middle| \hat{\mathbf{x}}_k; \mathbf{u}_k\right) \propto$$
$$\exp\left\{-\frac{1}{2}\mathbf{e}\left[\hat{\mathbf{x}}_k, \mathbf{u}_{k+1}, \hat{\mathbf{x}}_{k+1}\right]^\top \mathbf{Q}_{k+1}^{-1}\mathbf{e}\left[\hat{\mathbf{x}}_k, \mathbf{u}_{k+1}, \hat{\mathbf{x}}_{k+1}\right]\right\}$$

Step 1: Initializing the New Platform Vertex State

- We could use any logic to assign an initial value
- However, recall that optimizers have a “basin of convergence” so we’d like to initialize as close to true as possible
- Therefore, a common choice is to use the EKF linearization approximation for the mean,

$$\hat{\mathbf{x}}_{k+1} = \mathbf{f}\left[\hat{\mathbf{x}}_k, \mathbf{u}_{k+1}, 0\right]$$

Step 1: Observing Landmarks

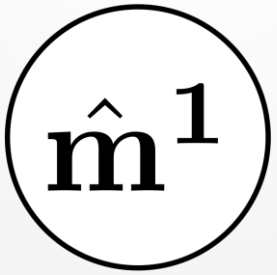


Step 1: Initializing Landmarks

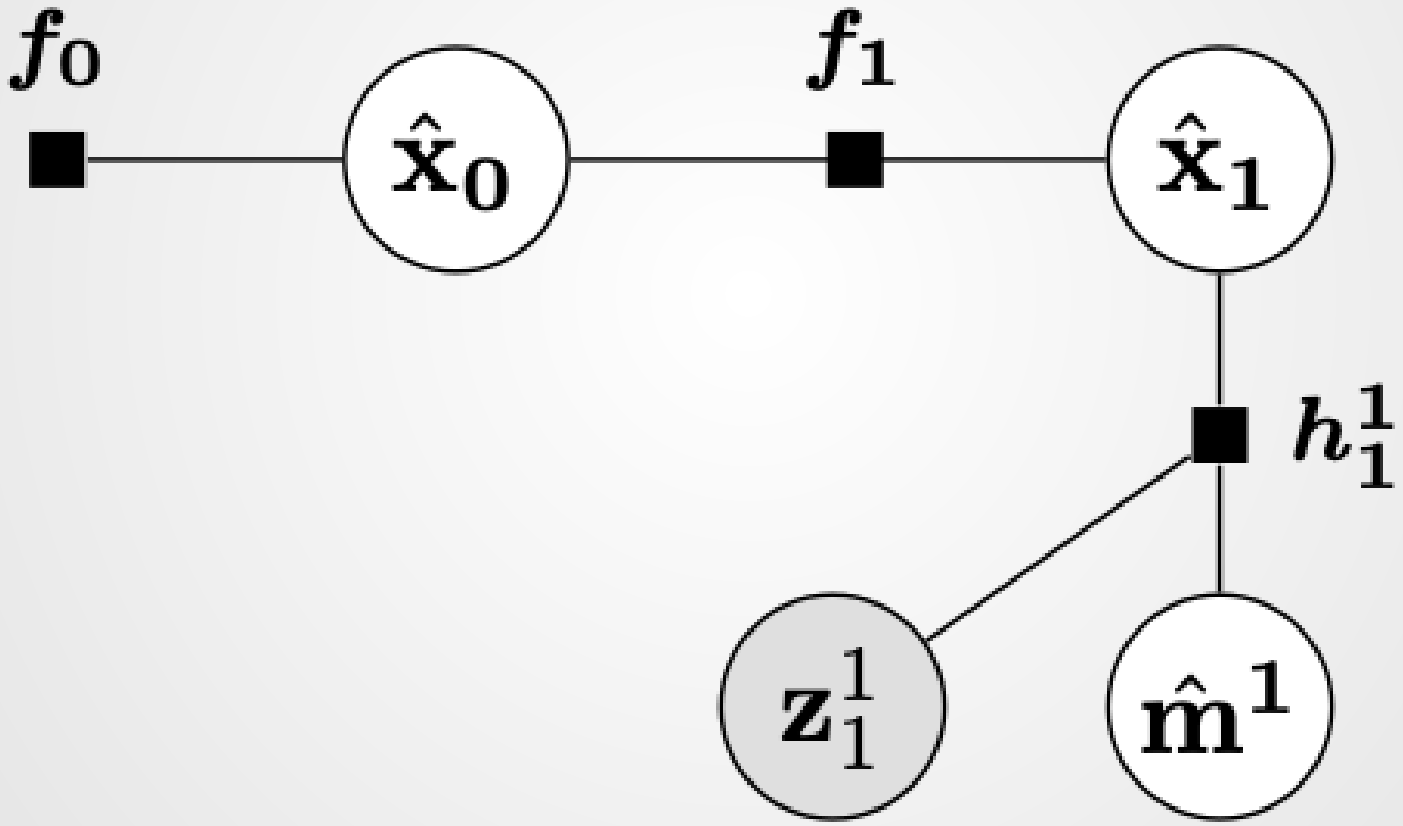
- The SLAM system now observes landmarks 1-4
- We will look at how to initialize one landmark – the procedure simply repeats for the other landmarks

Step 1: Initializing Landmarks

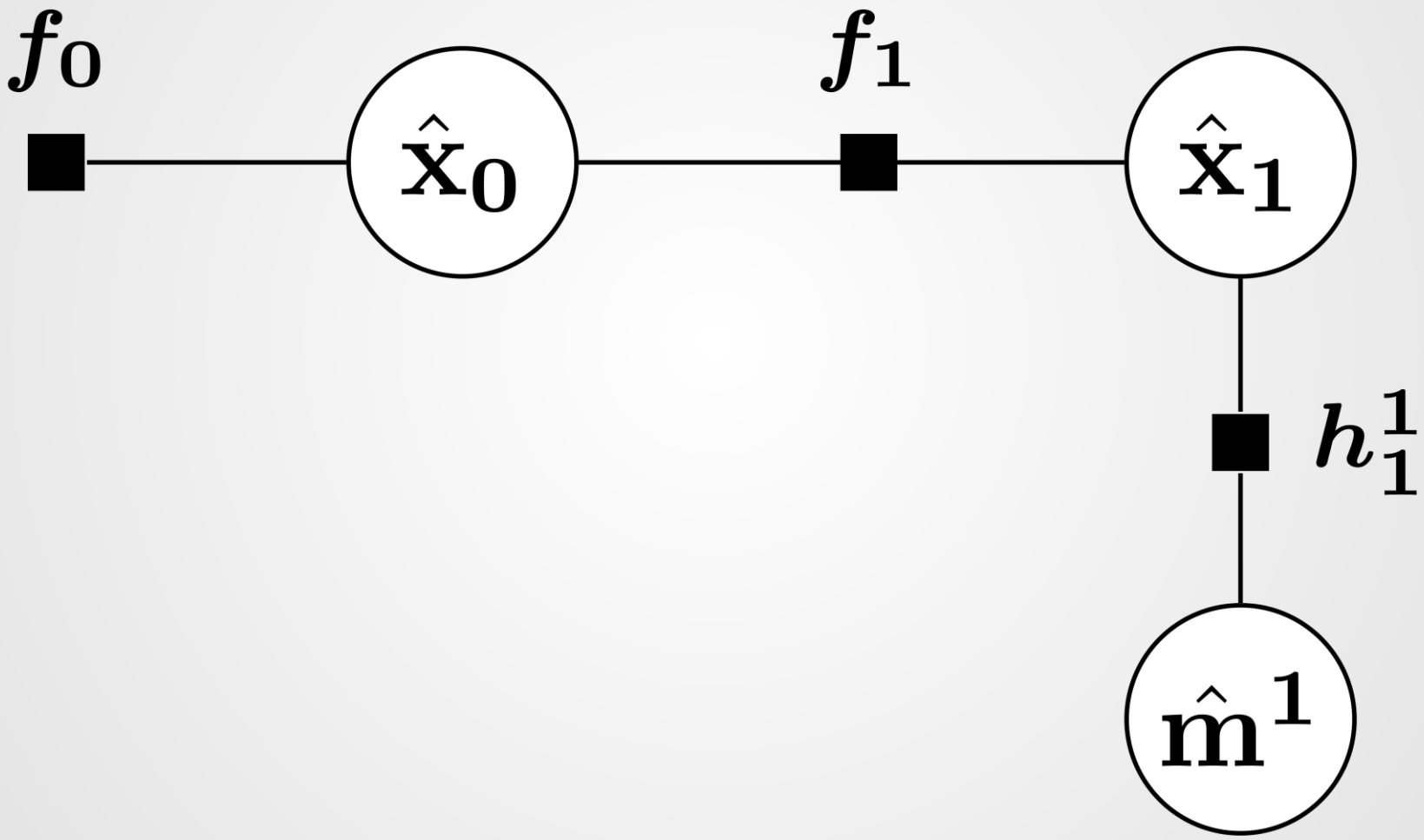
- The first thing is that we need to add a new vertex to the graph which corresponds to the landmark
- Since the landmark is constant over time, we only introduce a *single* vertex and use it for all time



Step 1: Graph Structure



Step 1: Simplified Graph Structure



Step 1: Landmark Initialization

- We now need to choose a factor which links the landmark observation with the platform state
- With the Kalman filter, we had to introduce the augmentation step to do it
- However, in the graph *augmentation is automatic* – every time you add a vertex with an edge to the graph, the back end works out all the correlations, etc. automatically

Step 1: Landmark Initialization Factor

- **Therefore, when initializing a landmark it turns out:**
  - **The factor is *the same type* that you use for a regular observation**
  - **The *only difference* is that you provide an initial value for the landmark estimate**

Step 1: Landmark Factor

- Recall the observation model which links the platform with the landmark state has the form

$$\mathbf{z}_{k+1}^{j} = \mathbf{h}\left[\mathbf{x}_{k+1}, \mathbf{m}^{i_j}, \mathbf{w}_{k+1}^{j}\right]$$

- Therefore, we can rearrange this to give

$$\mathbf{w}_{k+1}^{j} = \mathbf{l}\left[\mathbf{z}_{k+1}^{j}, \mathbf{x}_{k+1}, \mathbf{m}^{i_j}\right]$$

Step 1: Landmark Factor

- **Therefore, the observation likelihood factor has the form**

$$L\left(\hat{\mathbf{x}}_{k+1}, \hat{\mathbf{m}}^{i_j}; \mathbf{z}_{k+1}^{j}\right) \propto$$

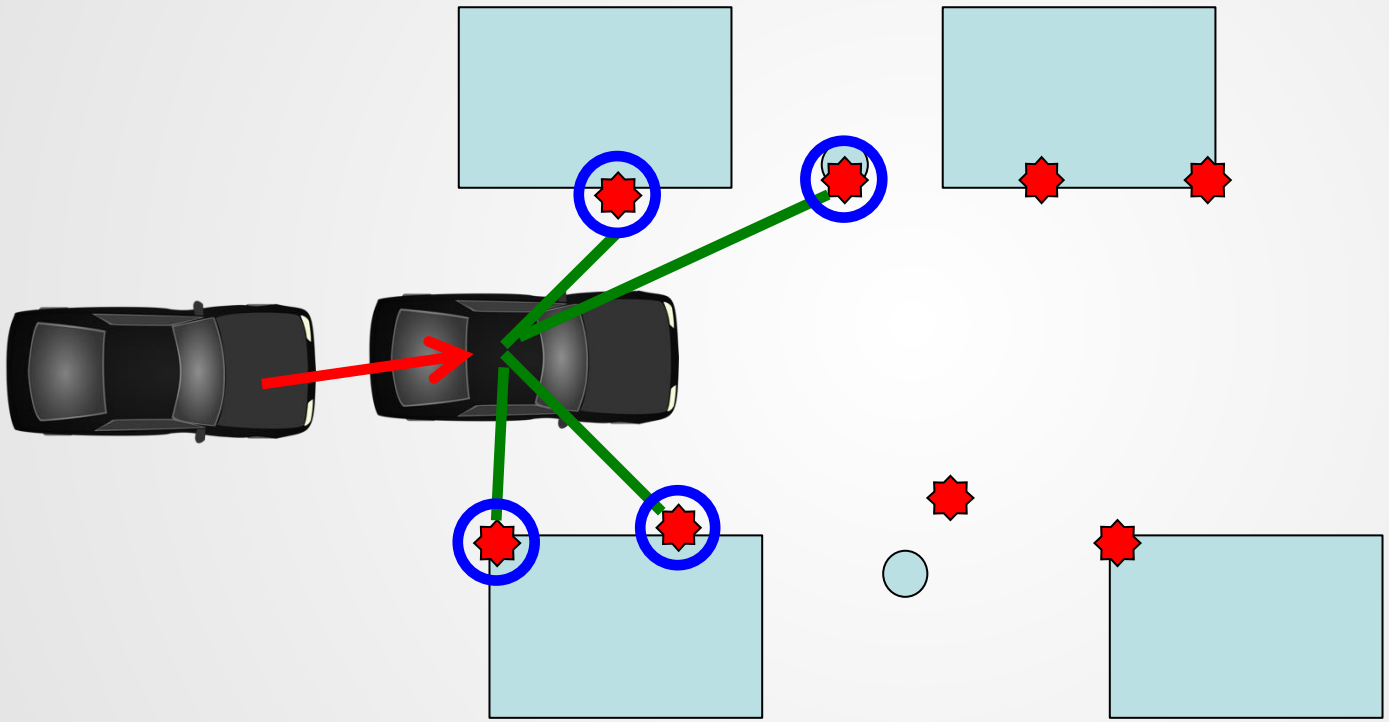
$$\exp\left\{\mathbf{l}\left[\mathbf{z}_{k+1}^{j}, \hat{\mathbf{x}}_{k+1}, \hat{\mathbf{m}}^{i_j}\right]^{\top} \mathbf{R}_{k+1}^{-1} \mathbf{l}\left[\mathbf{z}_{k+1}^{j}, \hat{\mathbf{x}}_{k+1}, \hat{\mathbf{m}}^{i_j}\right]\right\}$$

Step 1: Initializing the New Platform Vertex State

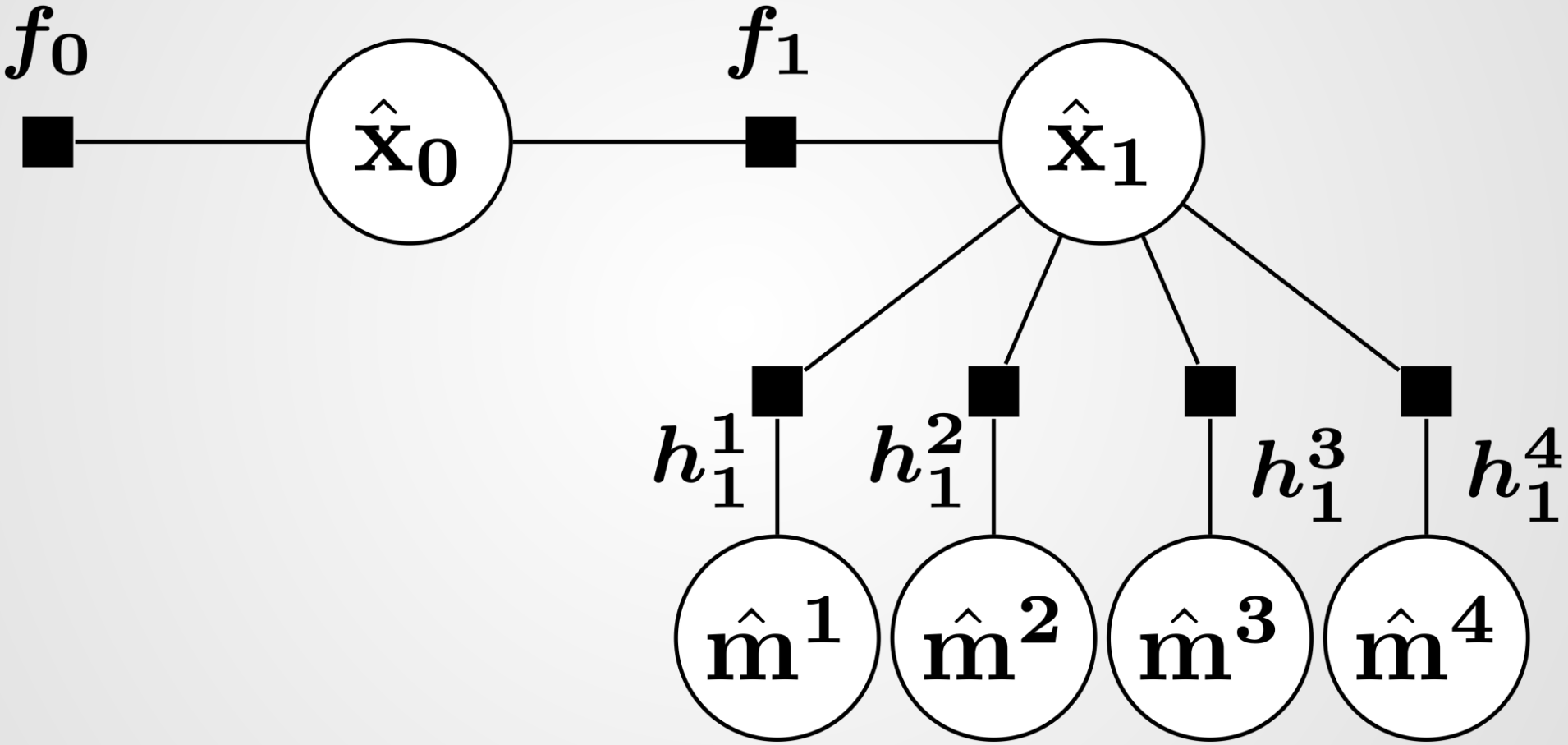
- We still need to initialize the landmark estimate
- As before, we could use any way to do this
- However, we want to try to initialize close to the true condition
- Therefore, we use the inverse observation model with the linearization assumption,

$$\hat{\mathrm{m}}^{i_j} = \mathrm{g}\left[\hat{\mathrm{x}}_{k+1}, \mathrm{z}^{j}_{k+1}, 0\right]$$

Step 1: Rest of the SLAM Observation



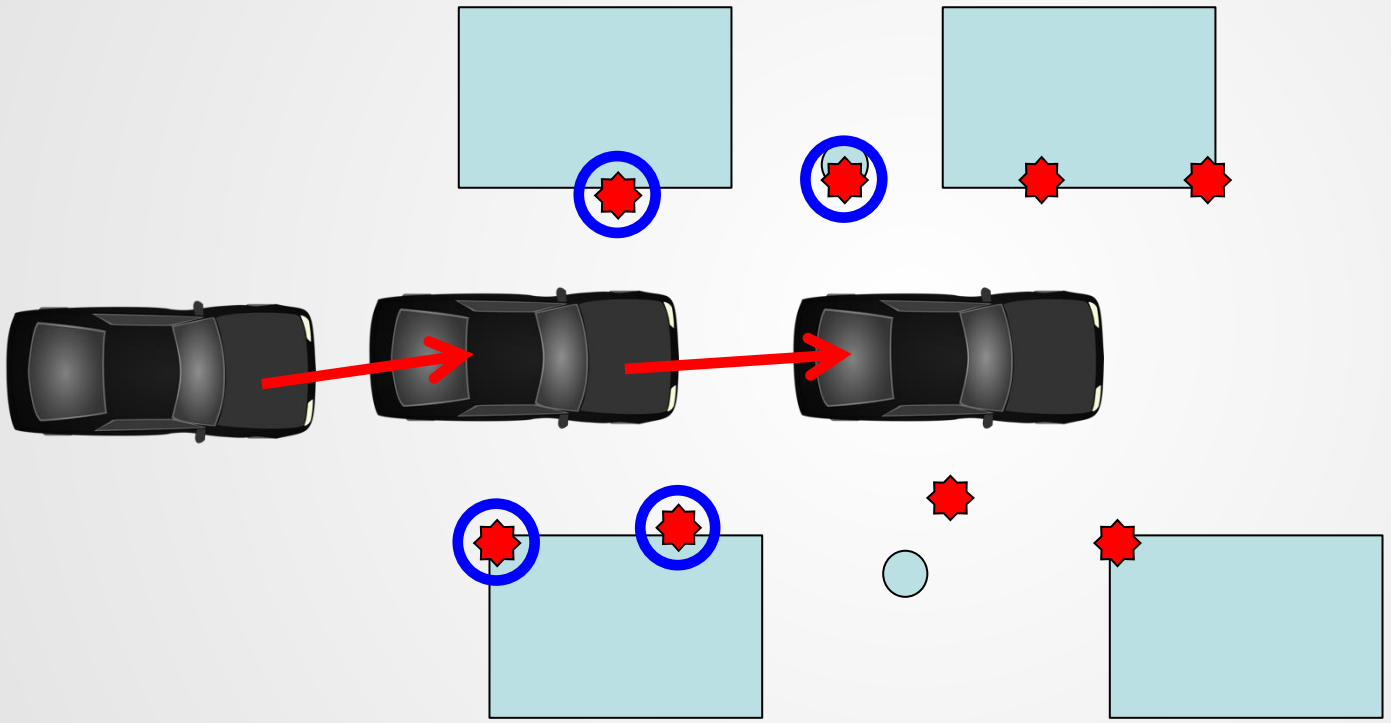
Step 1: Rest of the SLAM Observation



Factor Graph SLAM: The First Few Timesteps

Timestep	Action
~~0~~	~~Initialize~~
~~1~~	~~Predict and initalize landmarks 1-4~~
2	Predict and update landmarks 4, initialize landmarks 5-8

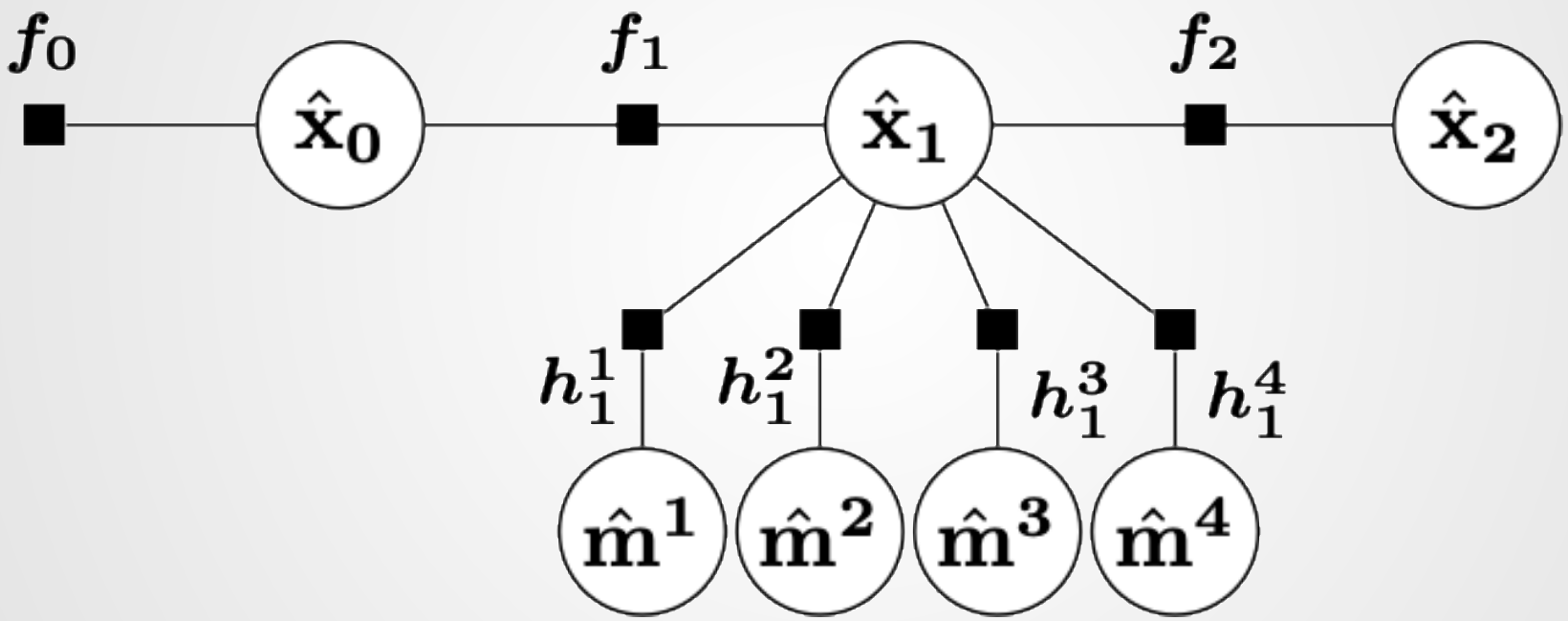
Step 2: Prediction



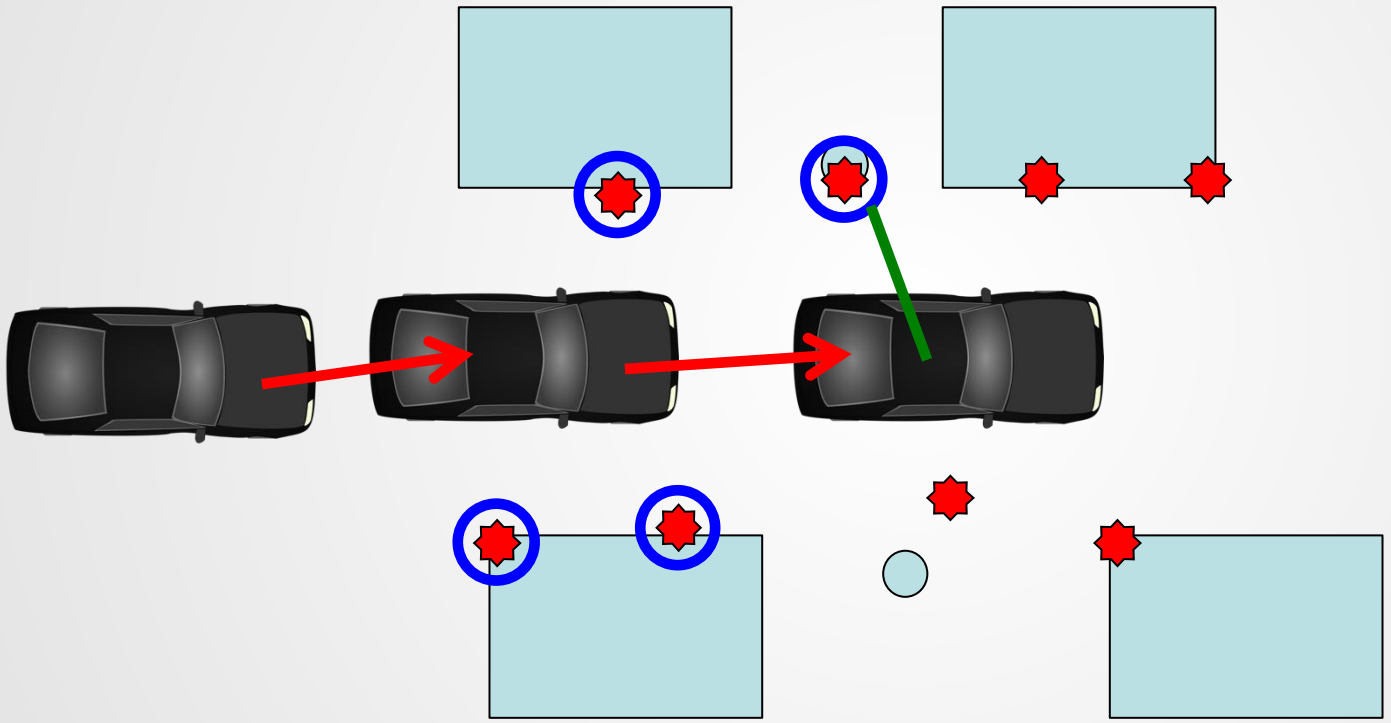
Step 2: Prediction

- We now want to jointly estimate the first three platform poses together with all the landmark estimates
- We do this by augmenting the graph with a new platform state and initializing its position just like we did before

Step 2: Prediction [REDRAW]



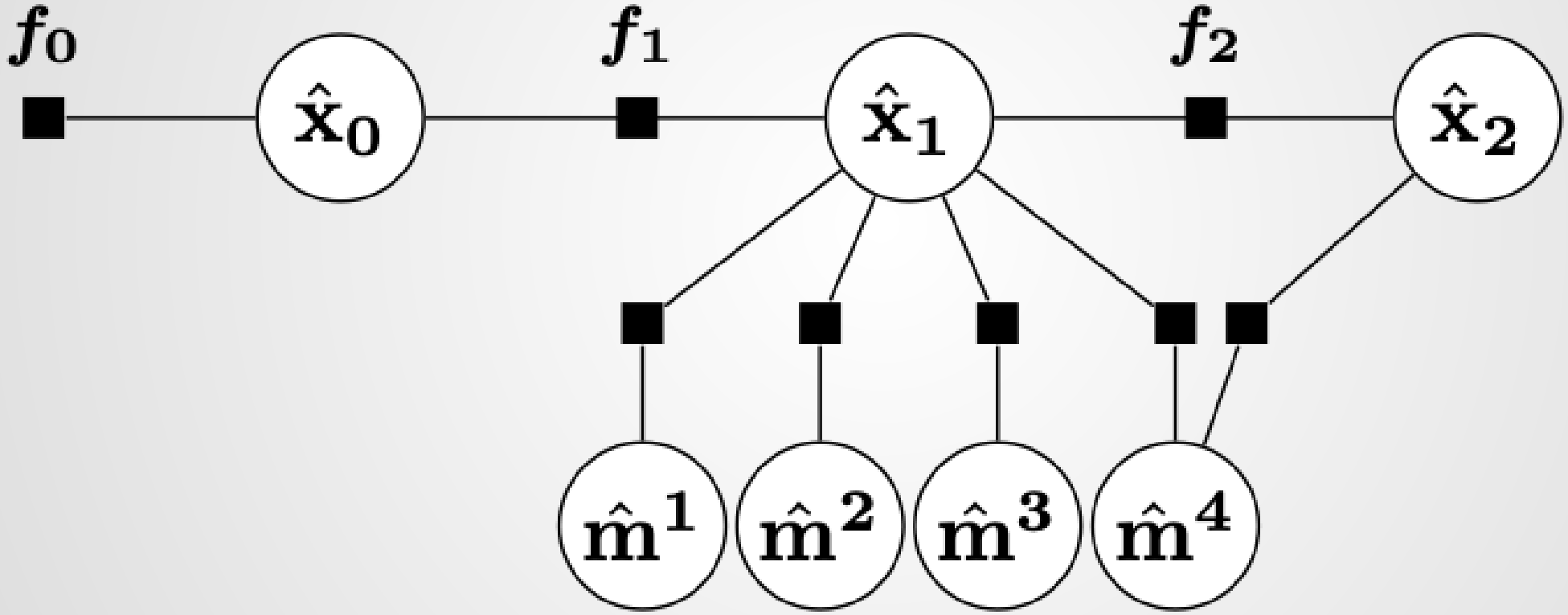
Step 2: Update



Step 2: Update

- We see one landmark ($\mathbf{m}^4$) a second time
- We model this by adding a *second edge* to $\mathbf{m}^4$
- This has the same form as the landmark observation edge we saw with initialization
- However, this time we do not set the estimate in the landmark vertex: the optimizer will update it

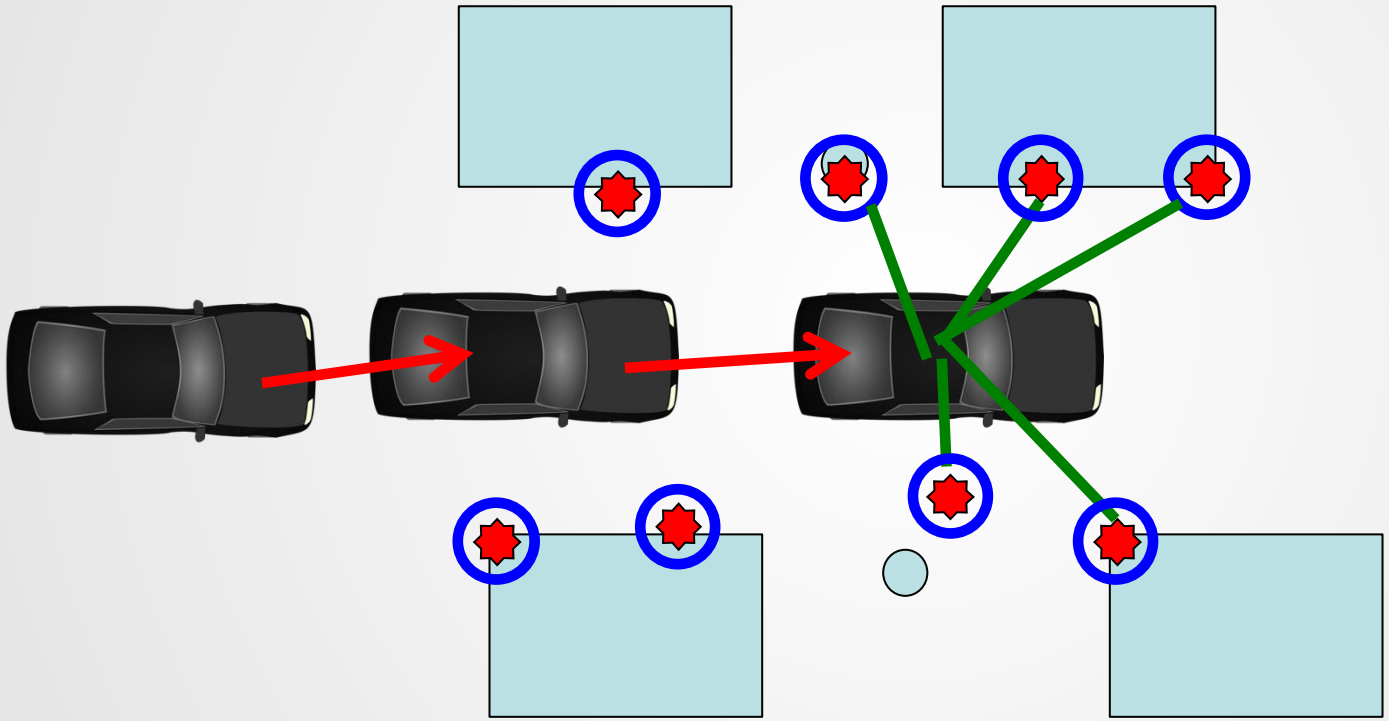
Step 2: Update



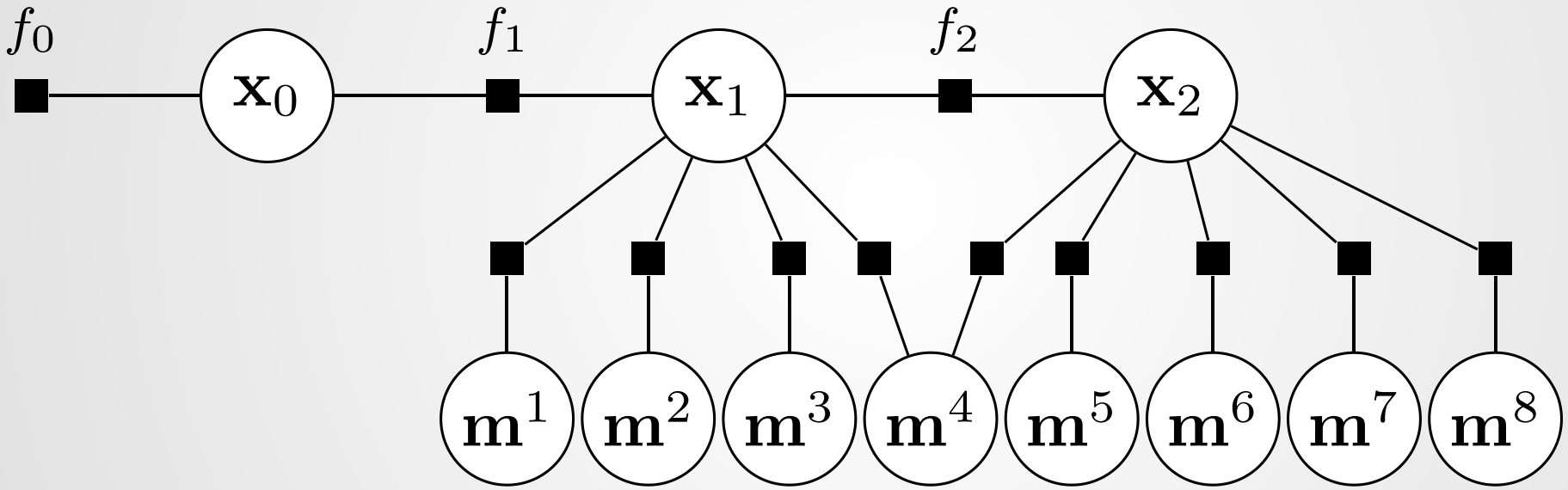
Step 2: Update

- We now need to initialize a new set of landmarks in the map
- We do the same process as before:
  - Create new vertices for the new landmarks
  - Set up the measurement edges
  - Initialize the landmark locations from the inverse observation model

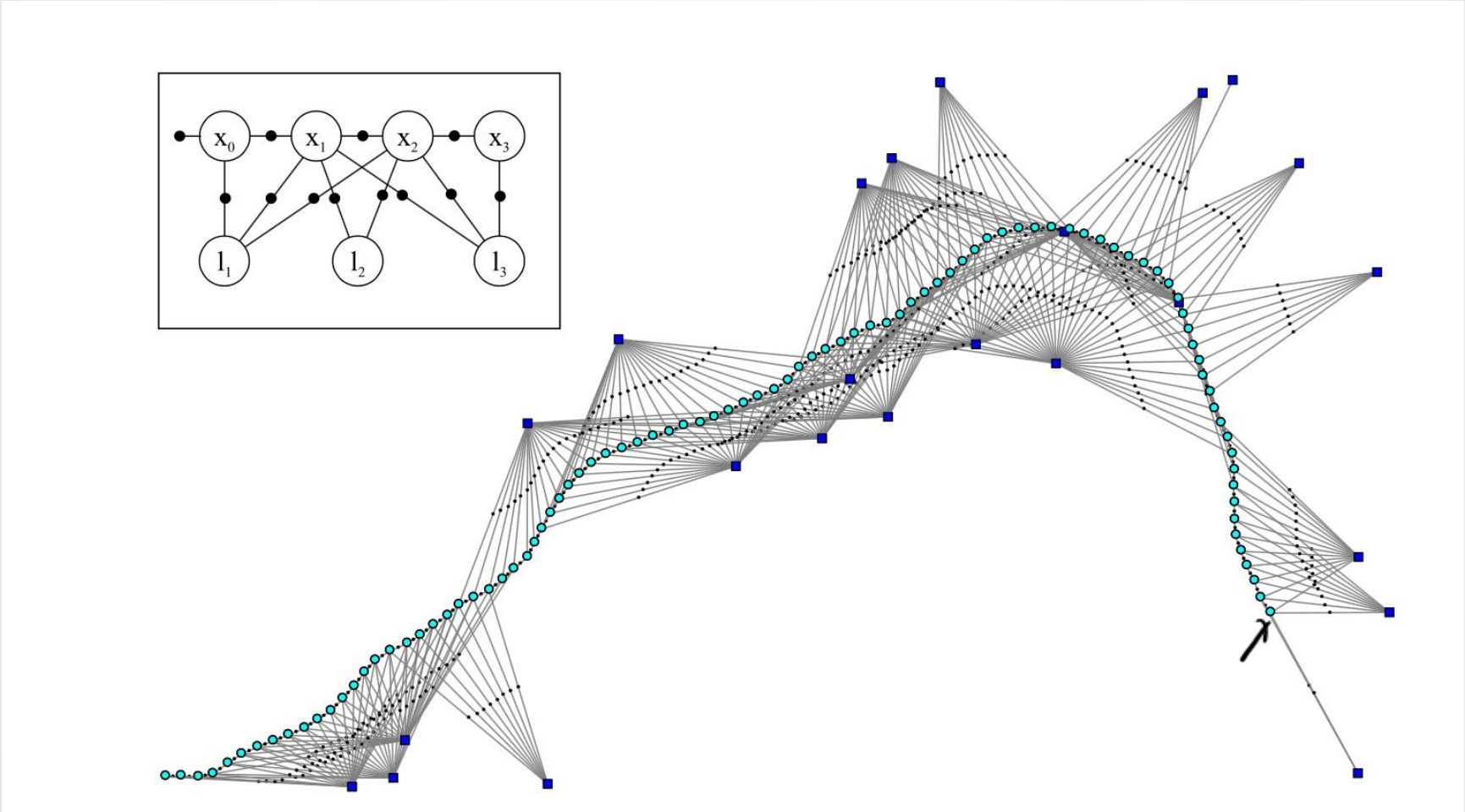
Step 2: Update



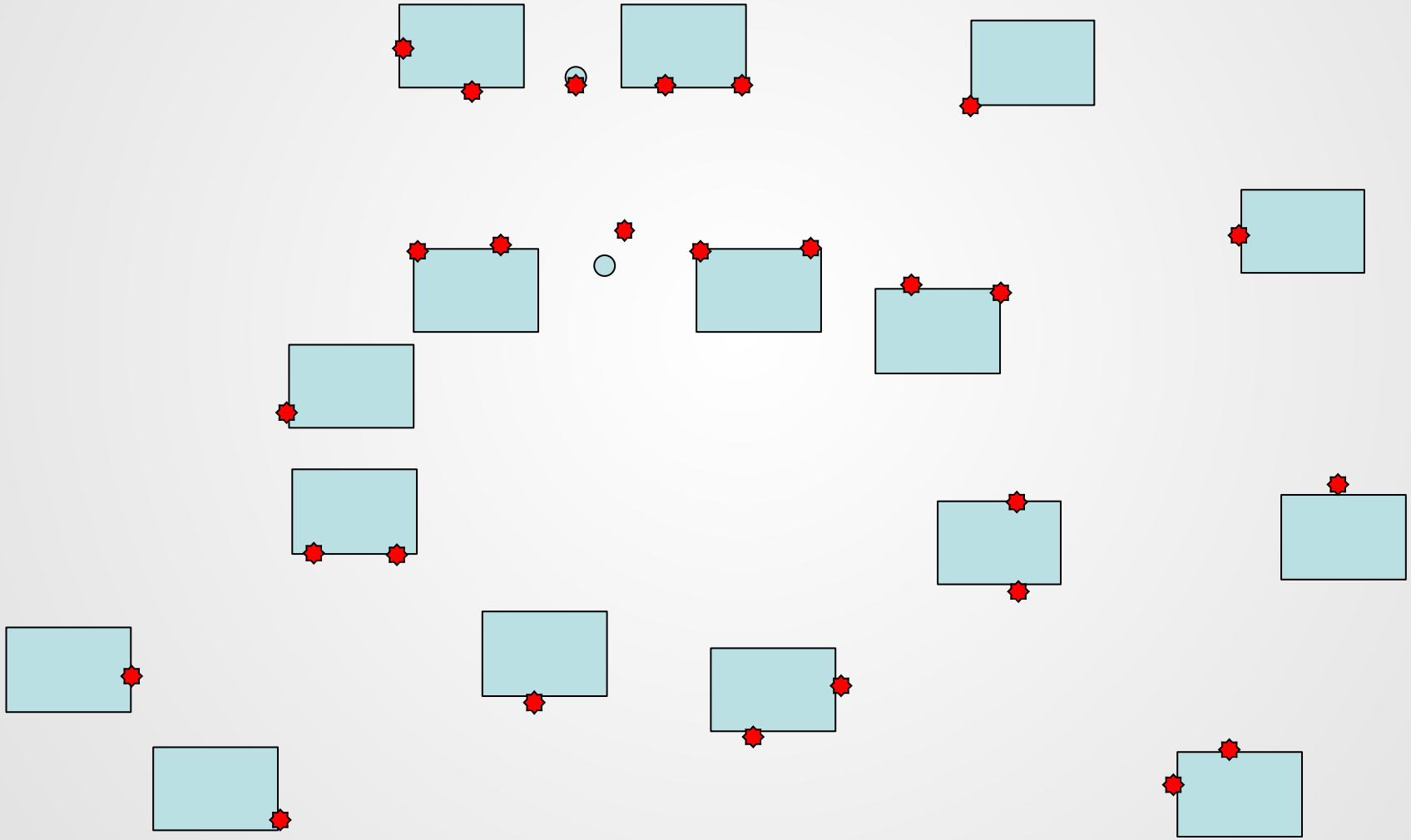
Step 2: Initialize Extra Landmarks



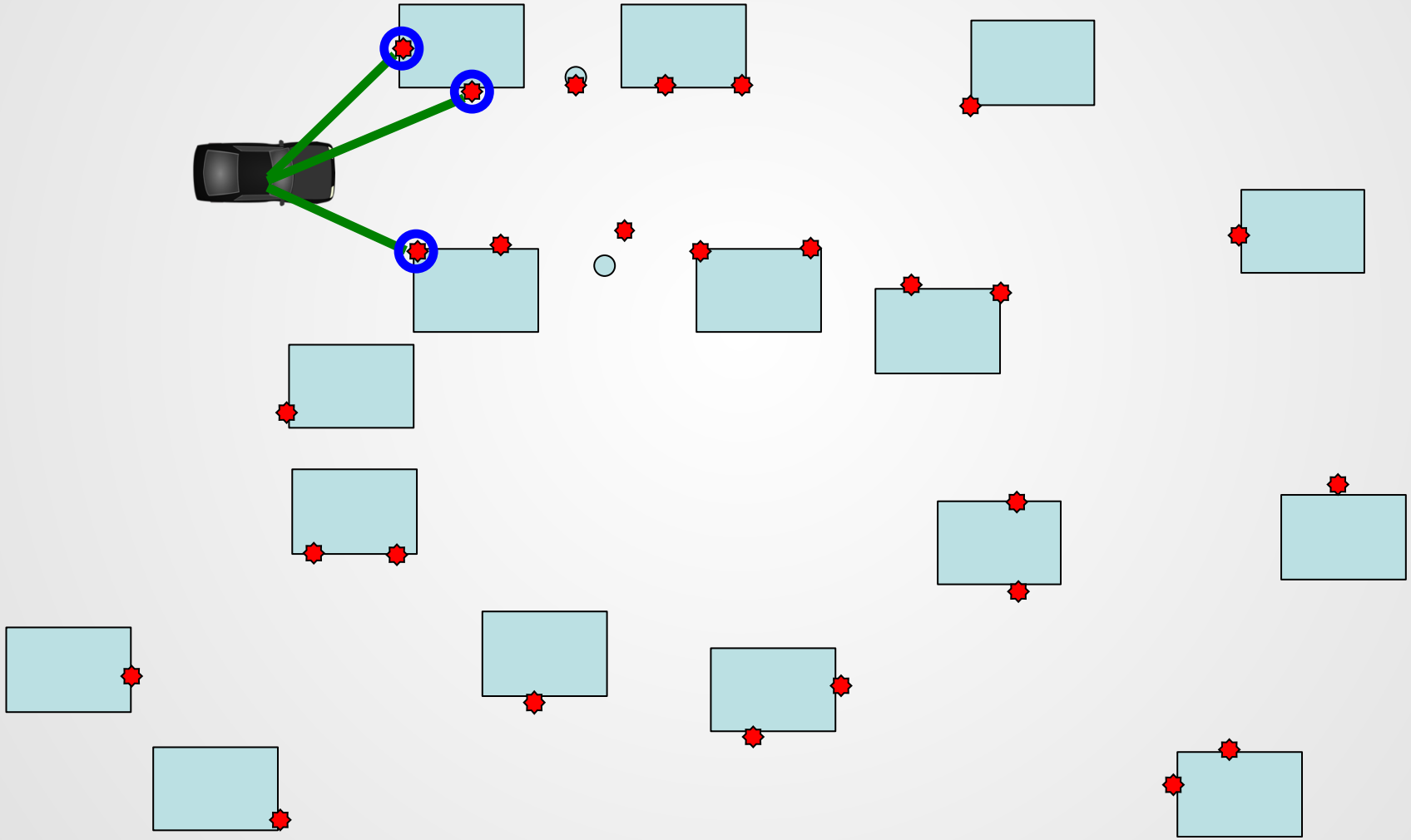
Bayes Net of Actual SLAM Example



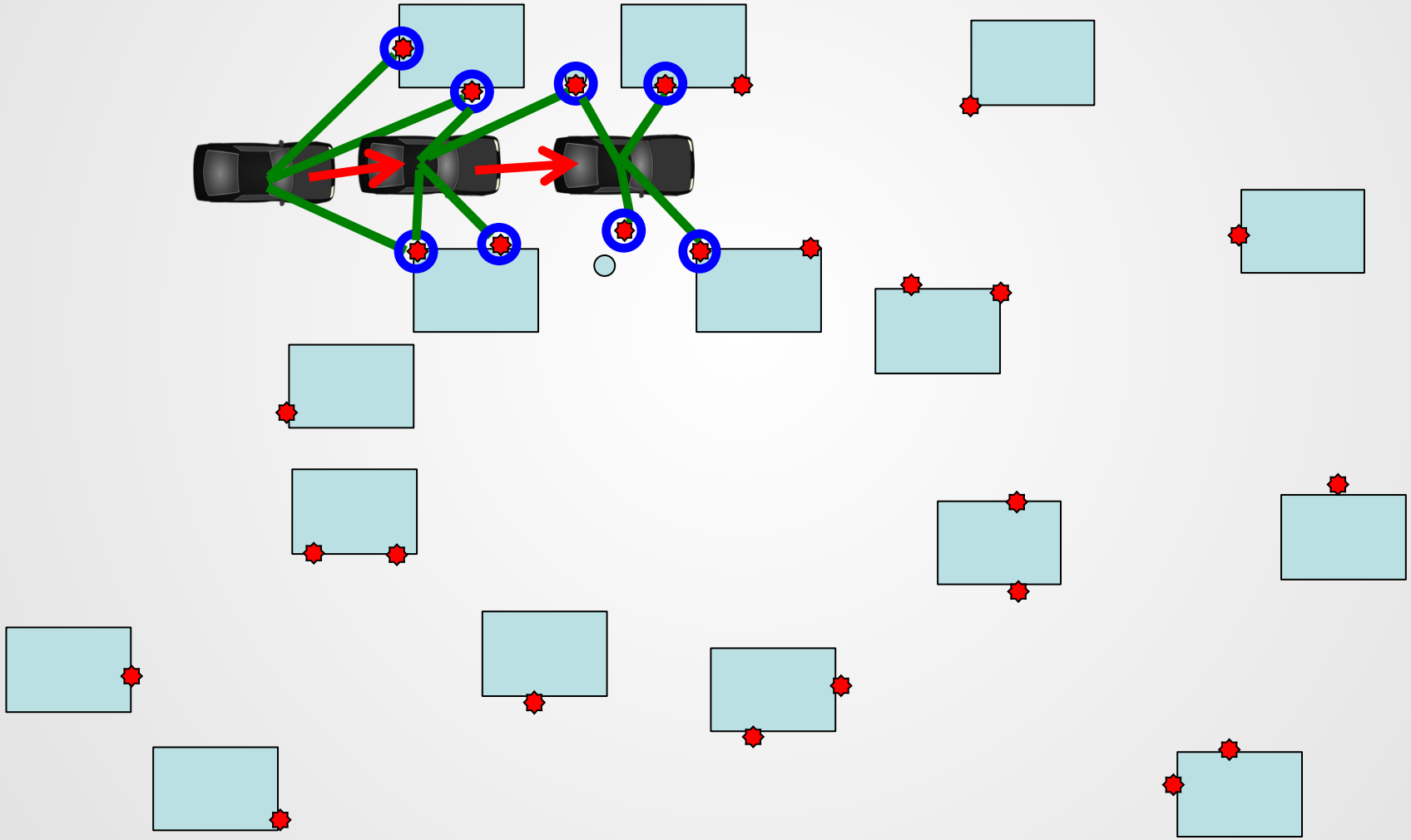
Illustrating Loop Closure



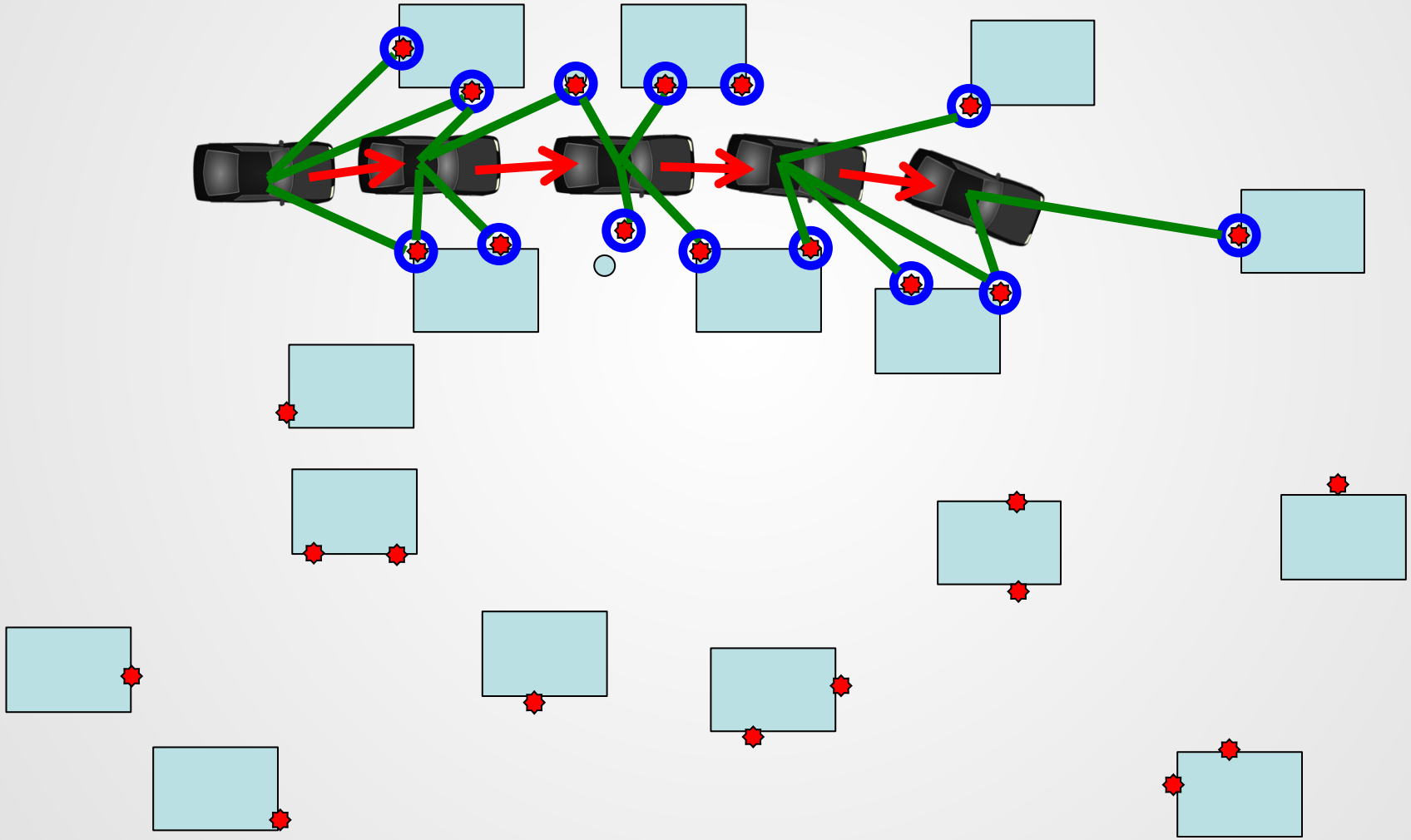
Illustrating Loop Closure



Illustrating Loop Closure



Illustrating Loop Closure



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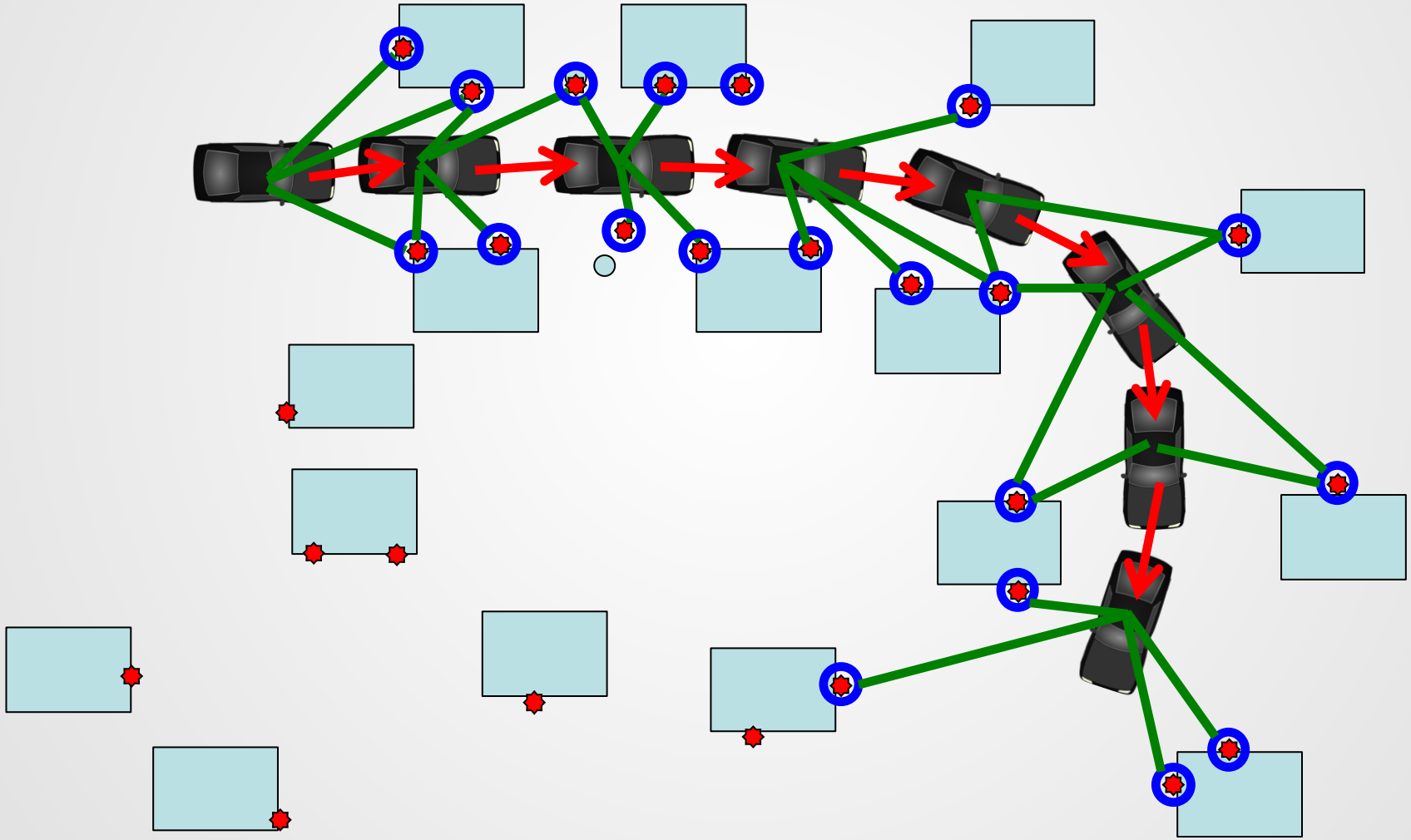
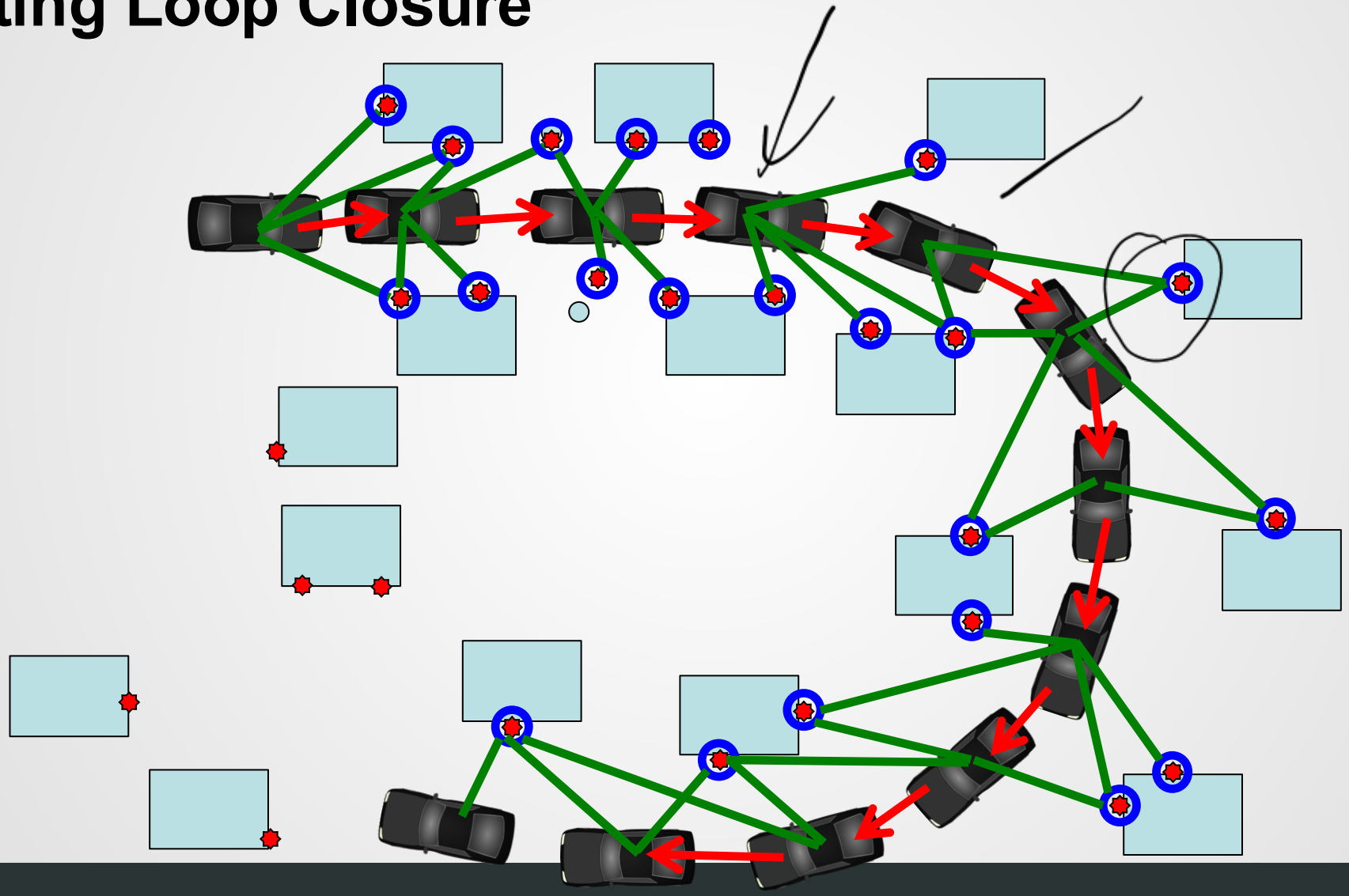
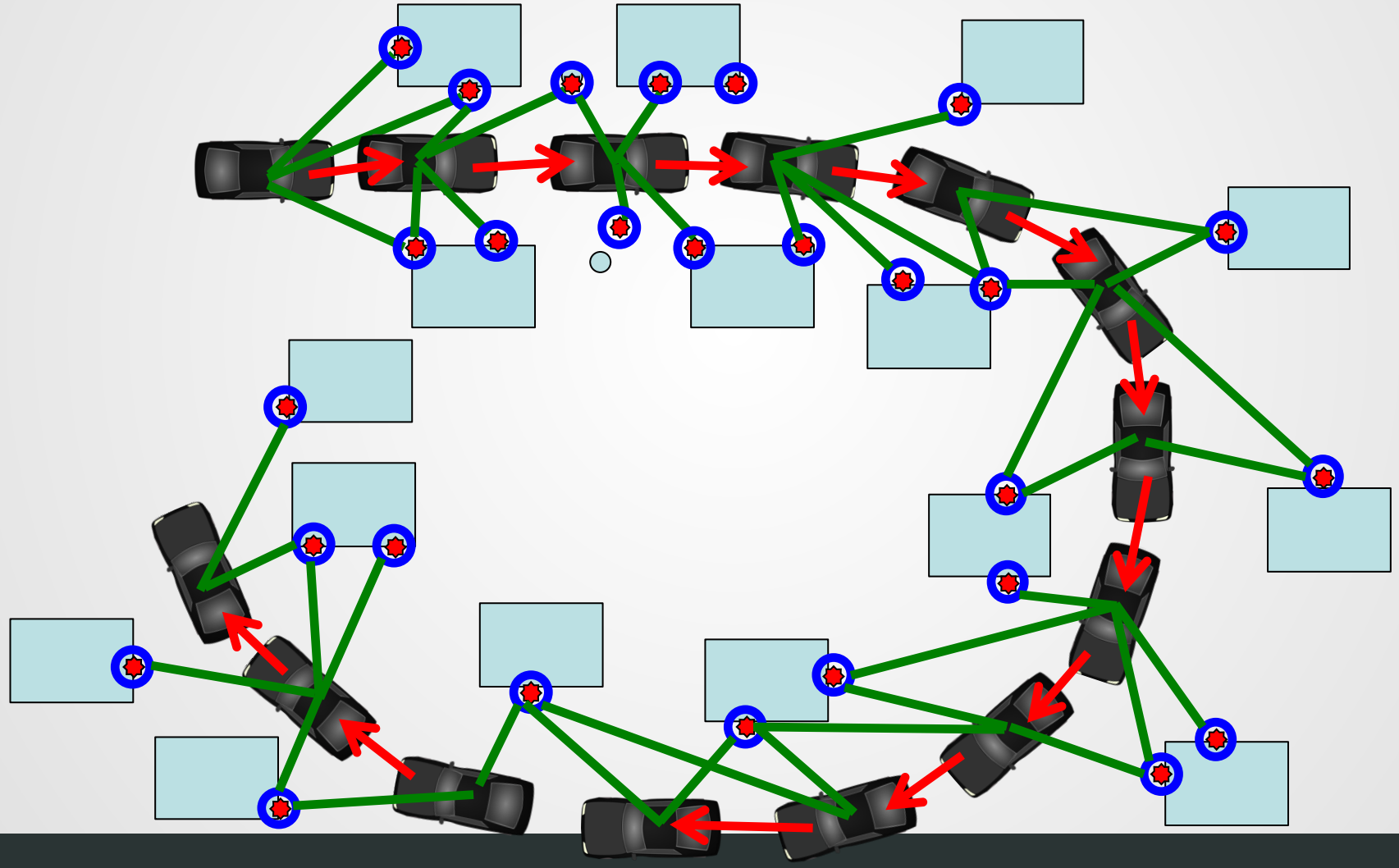


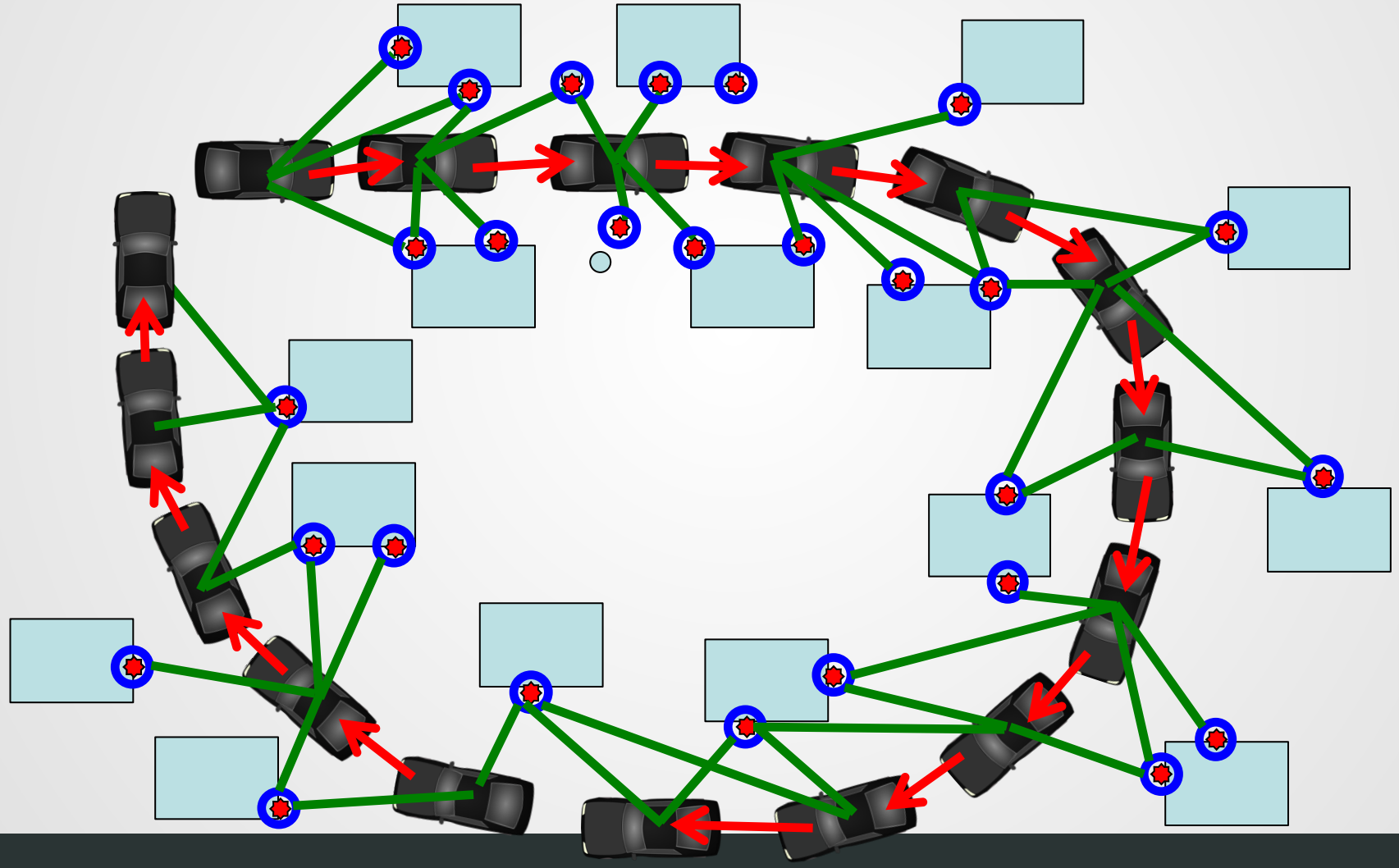
Illustrating Loop Closure



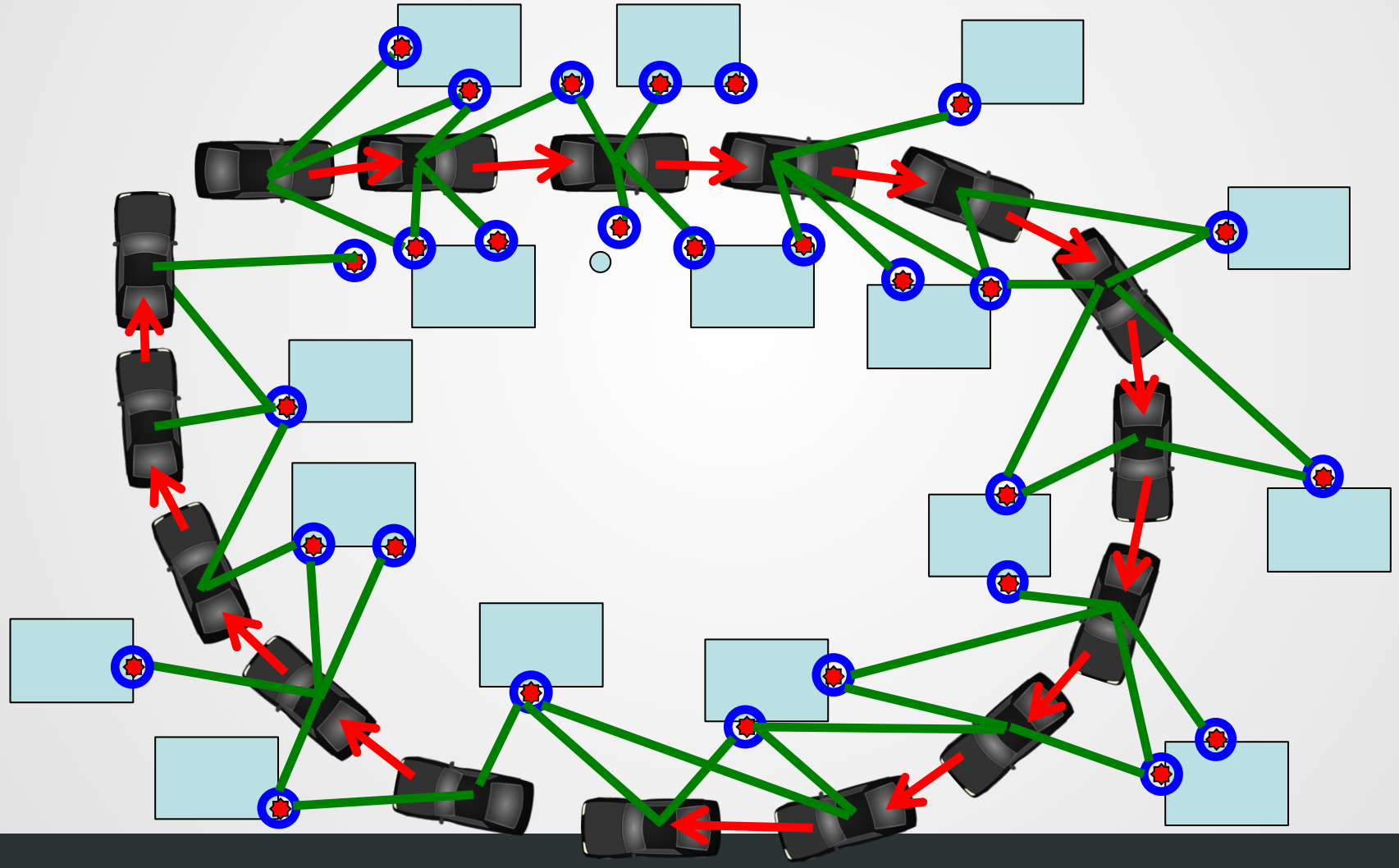
Illustrating Loop Closure



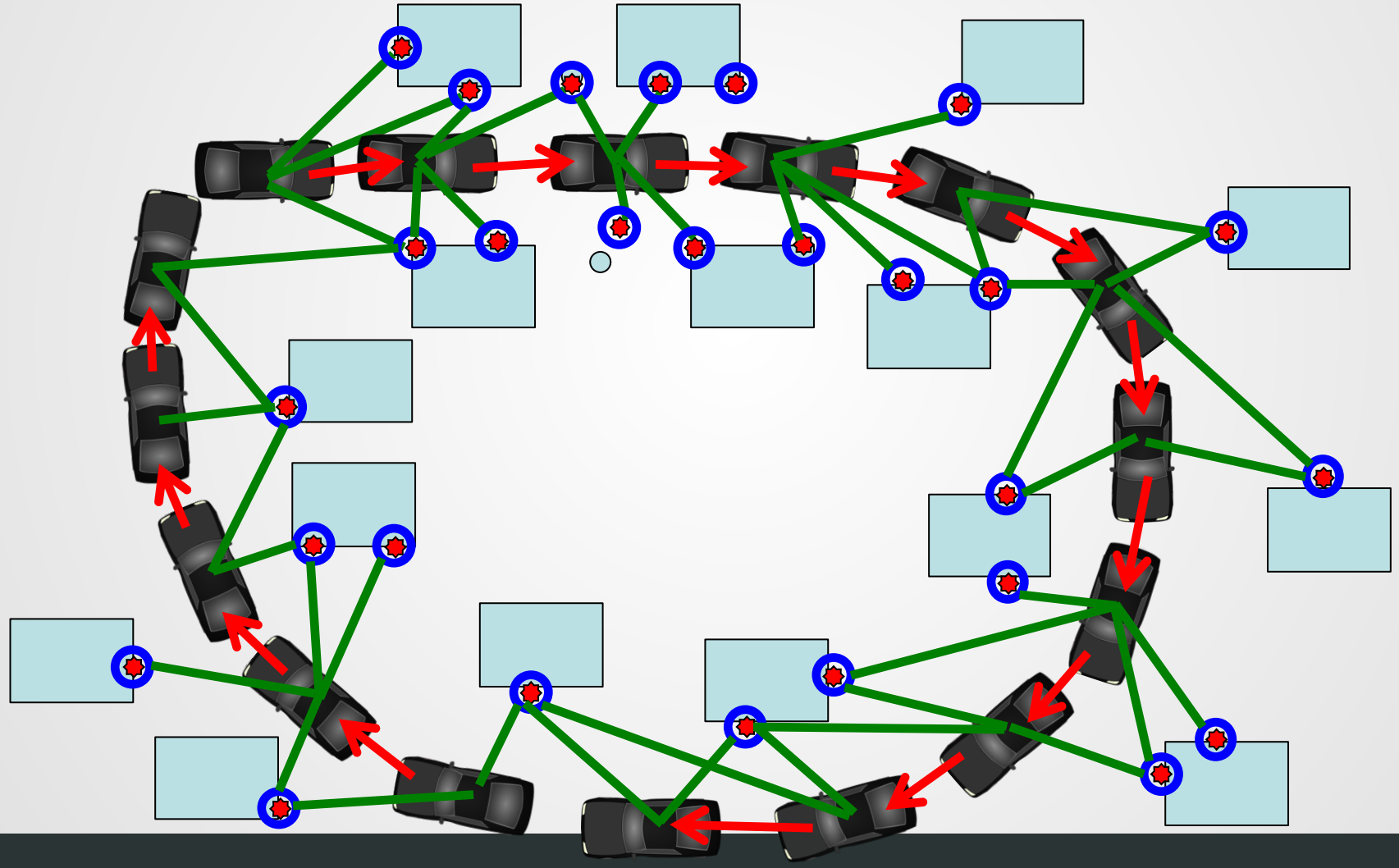
Illustrating Loop Closure



Illustrating Loop Closure



Illustrating Loop Closure



SLAM Examples

- *Formulating SLAM as a Graph Problem*
- **SLAM Examples**
- *Computational and Storage Costs*

Linear SLAM Example

- We'll use the example from Lab 02 in which we have a fully linear system
- The Kalman filter is *guaranteed to work*
- The Kalman filter is *optimal*
- We can see how to implement this and compare the performance directly with the Kalman filter

Platform State and Process Model

- The state of the platform is

$$\mathbf{x}_k = \begin{bmatrix} x_k \\ y_k \end{bmatrix}$$

Platform Process Model

- **The process model is**

$$\mathbf{x}_{k+1} = \mathbf{F}_{k+1}\mathbf{x}_k + \mathbf{B}_{k+1}\mathbf{u}_{k+1} + \mathbf{v}_{k+1}$$

Error for the Process Model Factor

- The error for the process model factor is given by

$$\mathbf{e}\left[\hat{\mathbf{x}}_{k+1}, \hat{\mathbf{x}}_k, \mathbf{u}_{k+1}\right] = \hat{\mathbf{x}}_{k+1} - \mathbf{F}_{k+1}\hat{\mathbf{x}}_k - \mathbf{B}_{k+1}\mathbf{u}_{k+1}$$

Landmark State

- **The landmark state is its position**

$$\mathbf{m}^i = \begin{bmatrix} x^i \\ y^i \end{bmatrix}$$

Landmark Observation Equation

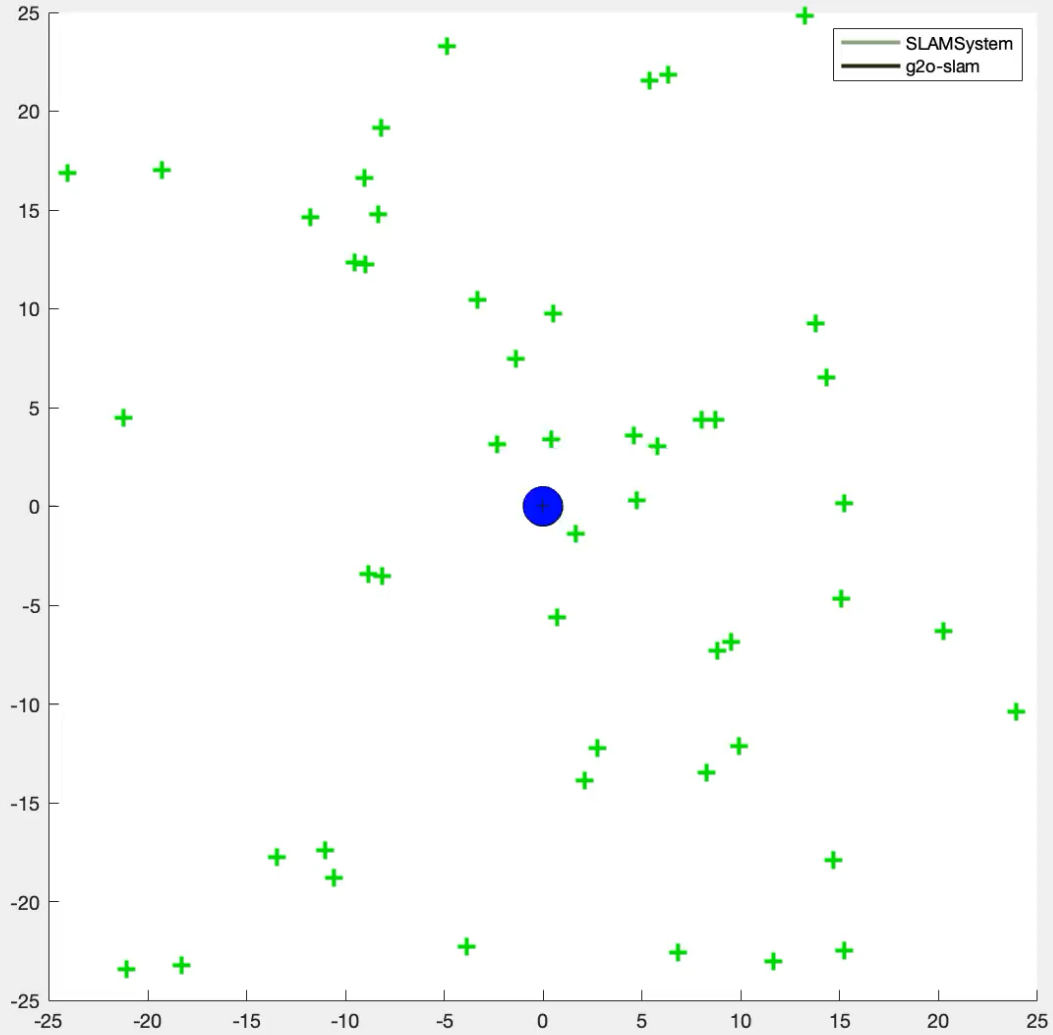
- **The landmark observation equation is**

$$\mathbf{z}_{k+1}^{j} = \mathbf{m}^{i_j} - \mathbf{x}_{k+1} + \mathbf{w}_{k+1}^{j}$$

Landmark Likelihood Factor

- **Therefore, the observation error term is**

$$\mathbf{l}\left[\mathbf{z}_{k+1}^{j}, \hat{\mathbf{m}}^{i_j}, \hat{\mathbf{x}}_{k+1}\right] = \mathbf{z}_{k+1}^{j} - \hat{\mathbf{m}}^{i_j} + \hat{\mathbf{x}}_{k+1}$$


SLAMSystem
g2o-slam
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25

Discussion

- We can see the two *are the same*
- The Kalman filter is optimal and so it's impossible to do better in this case
- Therefore, the fact the graph produces the same results shows that it's optimal as well, but it was put together in a very different way
- Things get interesting when the system is *nonlinear*

Nonlinear SLAM Example

- In this section we'll just show some results for a nonlinear system

- However, since this is central to CW01, most of the details will be omitted

Nonlinear SLAM Example

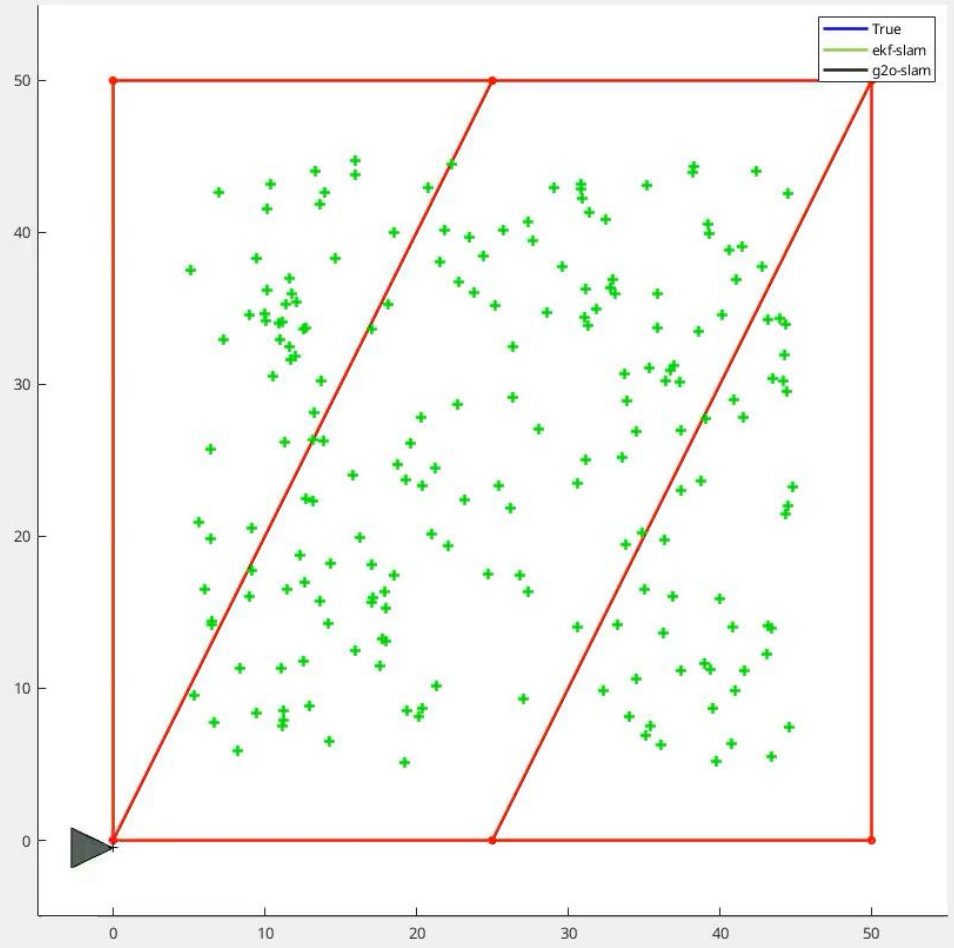
- As before, we have to introduce factors to:
  - Initialize the platform state
  - Predict the platform state over time
  - Handle platform-landmark observations

- In the coursework, you are provided with the code to initialize but you will need to write the rest yourself

Nonlinear SLAM Example

- We will also need to provide initialization code which produces numerical values for
  - To initialize the platform
  - To predict the next state of the platform
  - To initialize the platform estimate

- In the coursework, we provide the code to initialize the platform state, but you'll have to code the other bits yourself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0
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50

Computational and Storage Costs

- *Formulating SLAM as a Graph Problem*
- *SLAM Examples*
- Computational and Storage Costs

Storage and Computational Costs in the Kalman Filter

- In a Kalman filter, the storage and computational costs are dominated by the covariance matrix

- This has to store the covariance of the platform, all the landmarks, and all the cross correlations between them

Covariance

$$\mathbf{P}^{s}_{k|k} = \begin{bmatrix} \mathbf{P}^{\mathbf{xx}}_{k|k} & \mathbf{P}^{\mathbf{x}1}_{k|k} & \mathbf{P}^{\mathbf{x}2}_{k|k} & \mathbf{P}^{\mathbf{x}3}_{k|k} & \cdots & \mathbf{P}^{\mathbf{x}N_k}_{k|k} \\ \mathbf{P}^{1\mathbf{x}}_{k|k} & \mathbf{P}^{11}_{k|k} & \mathbf{P}^{12}_{k|k} & \mathbf{P}^{13}_{k|k} & \cdots & \mathbf{P}^{1N_k}_{k|k} \\ \mathbf{P}^{2\mathbf{x}}_{k|k} & \mathbf{P}^{21}_{k|k} & \mathbf{P}^{22}_{k|k} & \mathbf{P}^{23}_{k|k} & \cdots & \mathbf{P}^{2N_k}_{k|k} \\ \mathbf{P}^{3\mathbf{x}}_{k|k} & \mathbf{P}^{31}_{k|k} & \mathbf{P}^{32}_{k|k} & \mathbf{P}^{33}_{k|k} & \cdots & \mathbf{P}^{3N_k}_{k|k} \\ \vdots & \vdots & \ddots & \vdots & & \\ \mathbf{P}^{N_k\mathbf{x}}_{k|k} & \mathbf{P}^{N_k1}_{k|k} & \mathbf{P}^{N_k2}_{k|k} & \mathbf{P}^{N_k3}_{k|k} & \cdots & \mathbf{P}^{N_kN_k}_{k|k} \end{bmatrix}$$

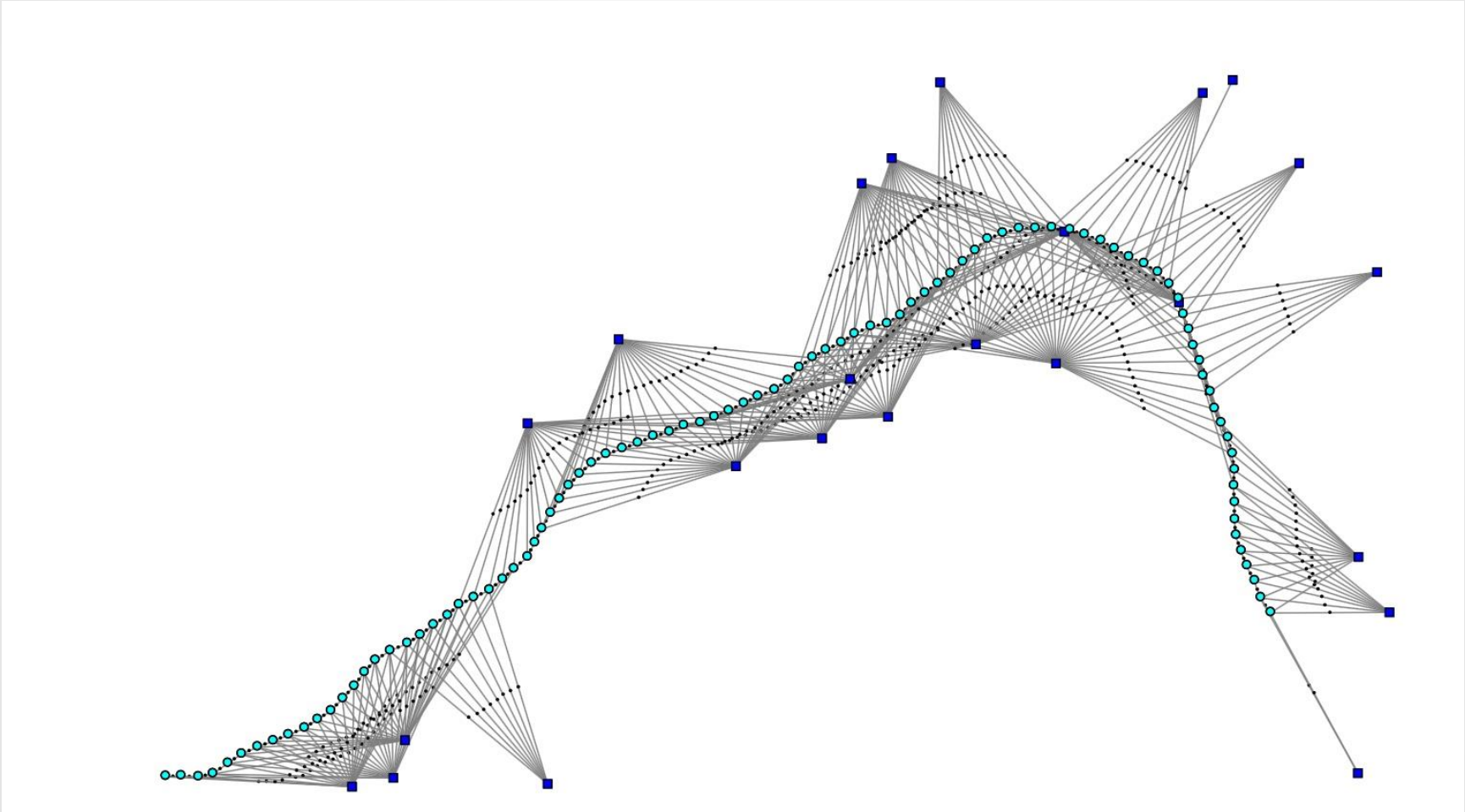
Storage and Computational Costs in the Factor Graph

- In the factor graph there are two main phases of operation:
  - Event recording
  - Graph optimization

Event Recording

- When we receive information we record it
- The way we do it is we instantiate additional vertices and edges
- Allocating each new element is constant time
- Storage grows *linearly* over time

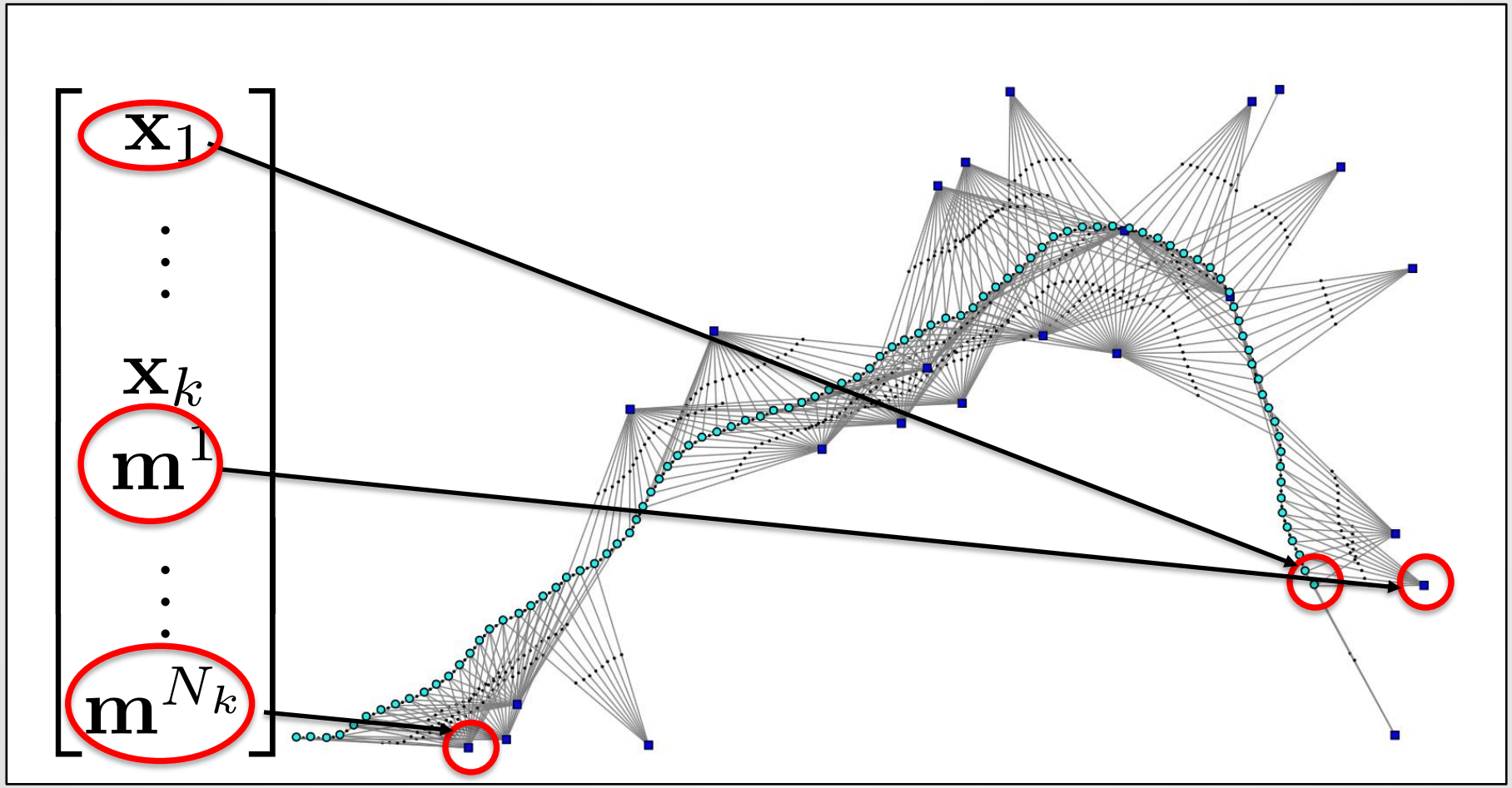
Growing Factor Graph



Graph Optimization

- We compute the MAP estimate of the graph
- To do this, we have to optimize
- Therefore, the optimization costs *dominate*

Assembling the System State



Optimization Step

- **The optimizers are iterative,**

$$\mathbf{X}_k^{i+1} = \mathbf{X}_k^i - \boldsymbol{\delta}\mathbf{X}_k^i$$

- **You can show (see the g2o paper) that the optimization step has the form**

$$\boldsymbol{\delta}\mathbf{X}_k^i = \mathbf{H}^{-1}\mathbf{b}$$

Hessian Matrix

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 c}{\partial x_1^2} & \frac{\partial^2 c}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 c}{\partial x_1 \partial x_s} \\ \frac{\partial^2 c}{\partial x_2 \partial x_1} & \frac{\partial^2 c}{\partial x_2^2} & \cdots & \frac{\partial^2 c}{\partial x_2 \partial x_s} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 c}{\partial x_s \partial x_1} & \frac{\partial^2 c}{\partial x_s \partial x_2} & \cdots & \frac{\partial^2 c}{\partial x_s^2} \end{bmatrix}$$

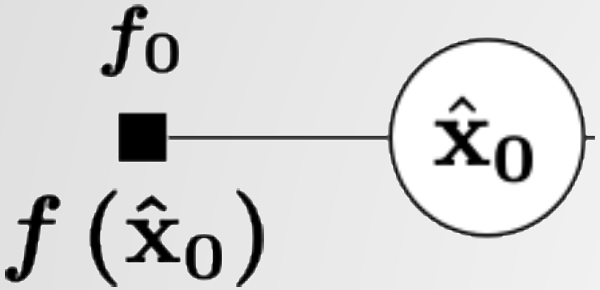
Using the Hessian

- It turns out that the inverse of the Hessian has a very easy form to write down
- Therefore, we compute the inverse of the Hessian directly
- However, the computational and storage costs depend on the number of entries in the Hessian
- Because Hessian encodes the structure of the graph, it is *sparse* – most of the terms are non-zero

Structure of the Hessian

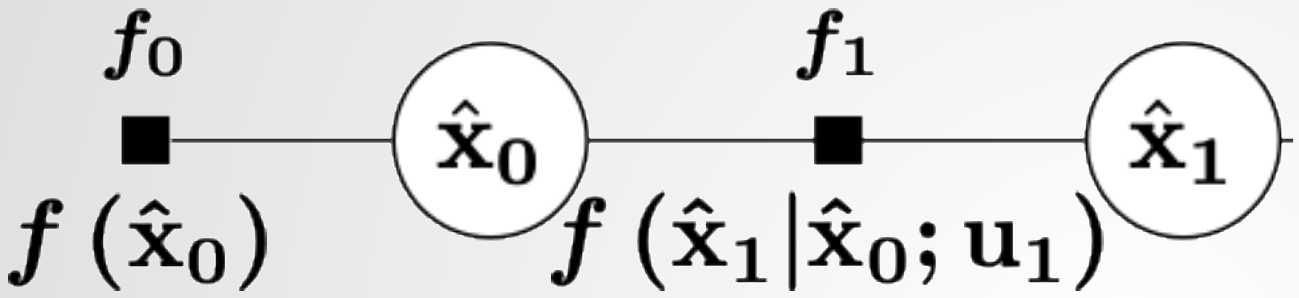
Row \ Col	$\hat{\mathbf{x}}_0$	$\cdots$	$\cdots$	$\cdots$
$\hat{\mathbf{x}}_0$				
$\vdots$				
$\vdots$				
$\vdots$				

Structure of the Hessian for Prediction Problems



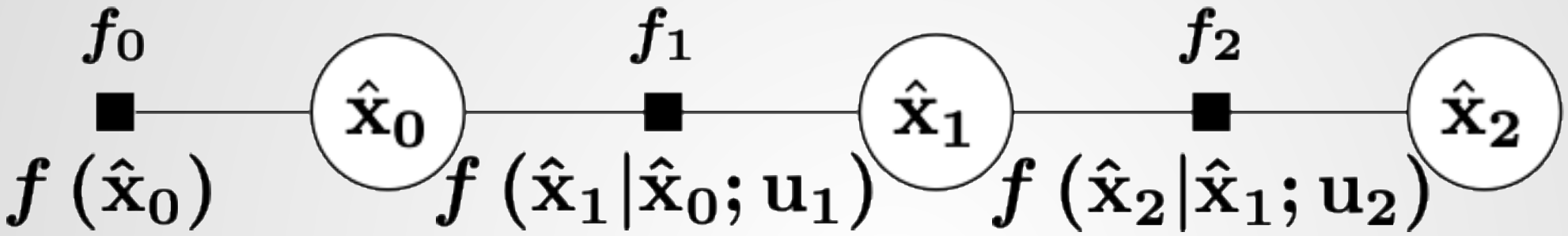
Row \ Col	$\hat{\mathbf{x}}_0$
$\hat{\mathbf{x}}_0$	

Structure of the Hessian for Prediction Problems



Row \ Col	$\hat{\mathbf{x}}_0$	$\hat{\mathbf{x}}_1$
$\hat{\mathbf{x}}_0$		
$\hat{\mathbf{x}}_1$		

Structure of the Hessian for Prediction Problems



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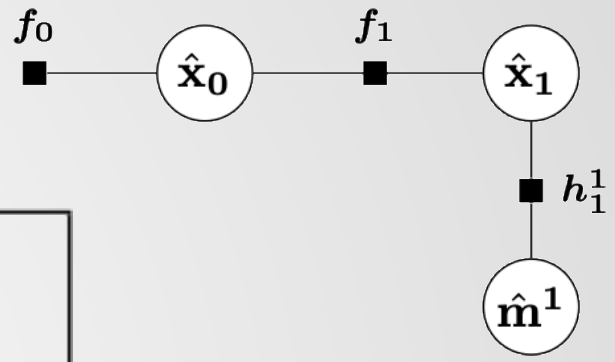
$\hat{\mathbf{x}}_0$			
$\hat{\mathbf{x}}_1$			
$\hat{\mathbf{x}}_2$			

Structure of the Hessian for Prediction Problems

Row \ Col	$\hat{\mathbf{x}}_0$	$\hat{\mathbf{x}}_1$	$\hat{\mathbf{x}}_2$	$\hat{\mathbf{x}}_3$	$\hat{\mathbf{x}}_4$	$\hat{\mathbf{x}}_5$
$\hat{\mathbf{x}}_0$						
$\hat{\mathbf{x}}_1$						
$\hat{\mathbf{x}}_2$						
$\hat{\mathbf{x}}_3$						
$\hat{\mathbf{x}}_4$						
$\hat{\mathbf{x}}_5$						

Structure of the Hessian with Landmarks



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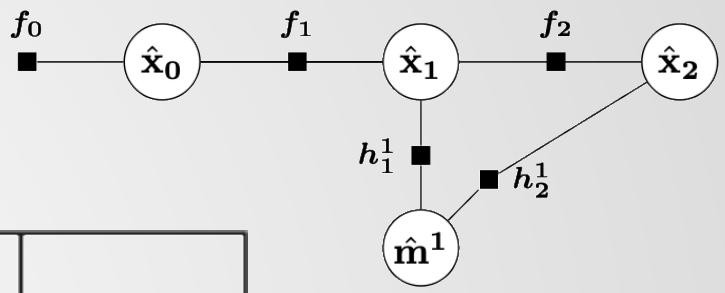
$\hat{\mathbf{x}}_0$			
$\hat{\mathbf{x}}_1$			
$\hat{\mathbf{m}}^1$			

Structure of the Hessian with Landmarks



Row \ Col	$\hat{\mathbf{x}}_0$	$\hat{\mathbf{x}}_1$	$\hat{\mathbf{m}}^1$	$\hat{\mathbf{x}}_2$
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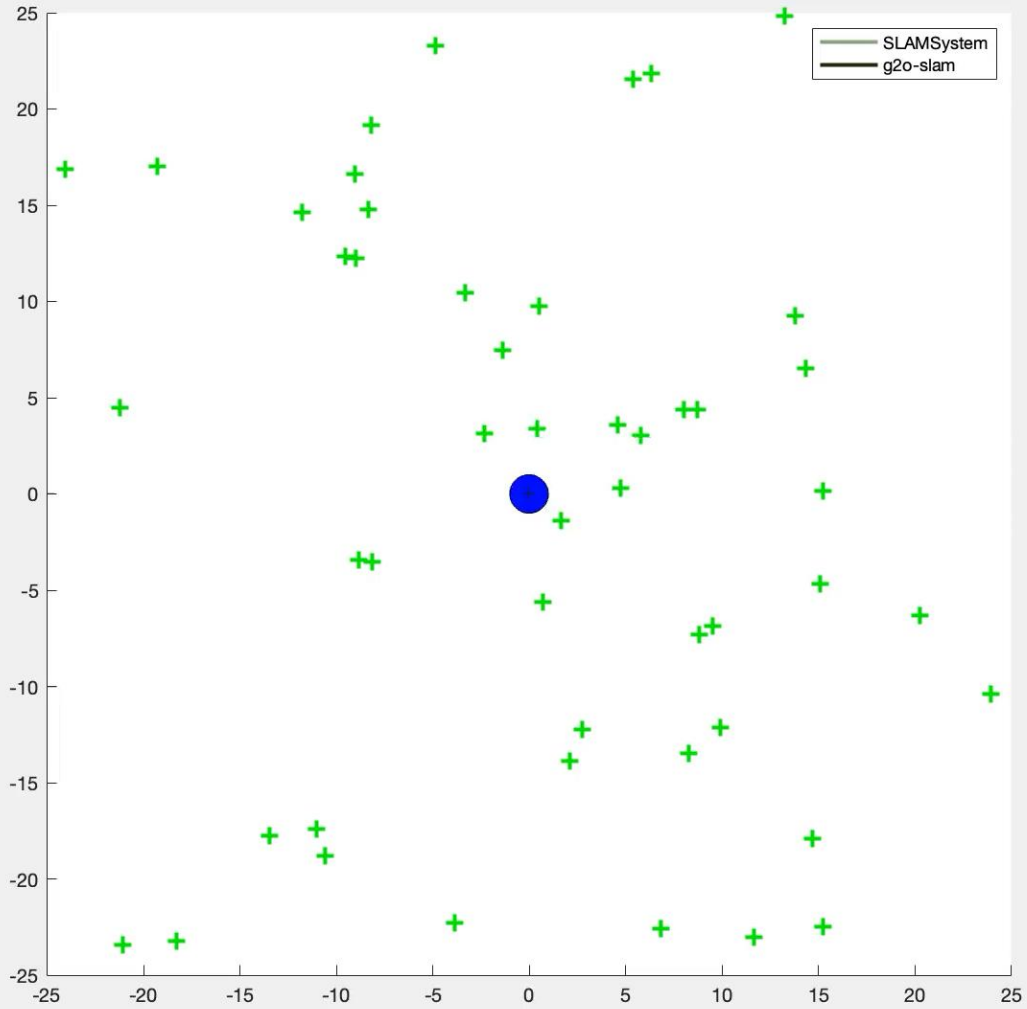
$\hat{\mathbf{x}}_1$				
$\hat{\mathbf{m}}^1$				
$\hat{\mathbf{x}}_2$				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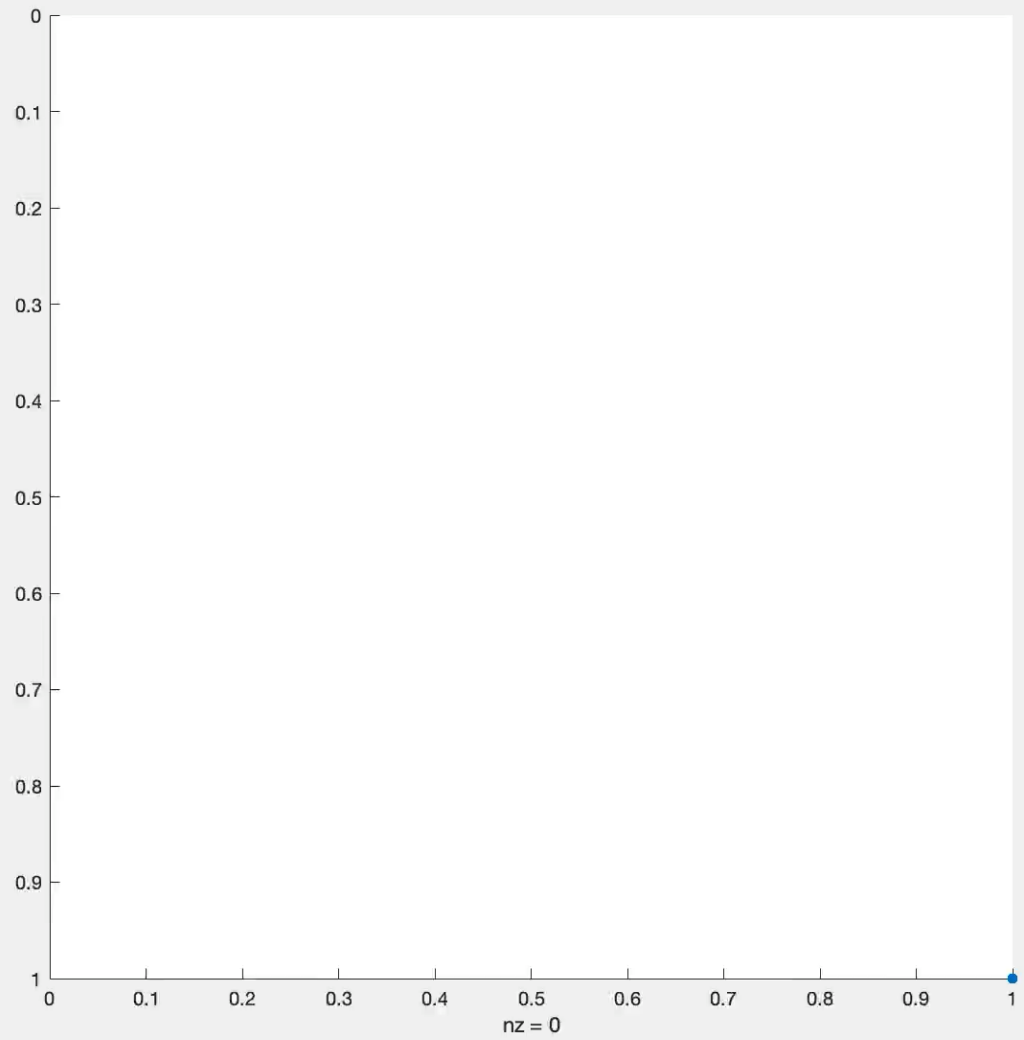
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0.7
0.8
0.9
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0.4
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0.7
0.8
0.9
1
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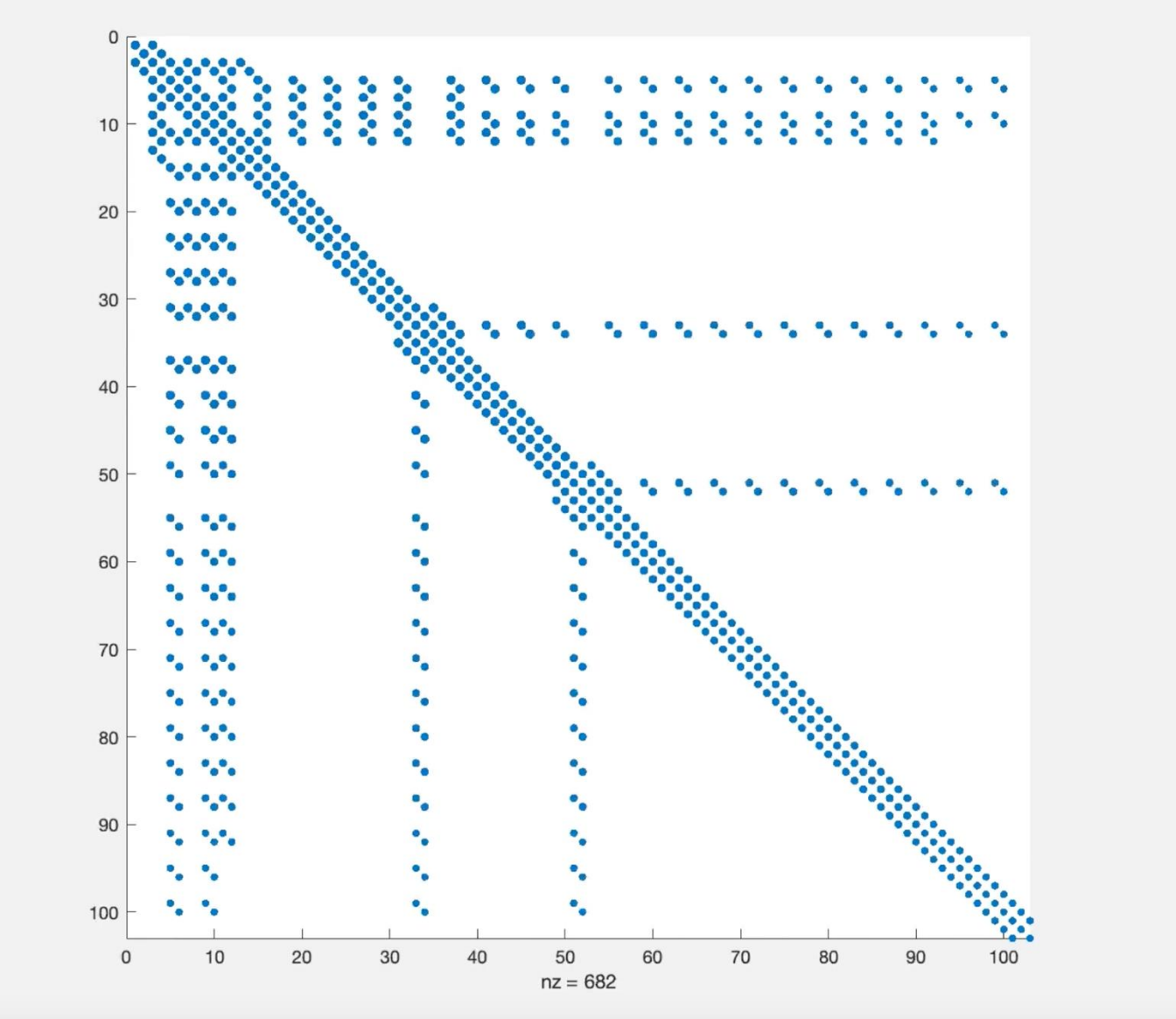
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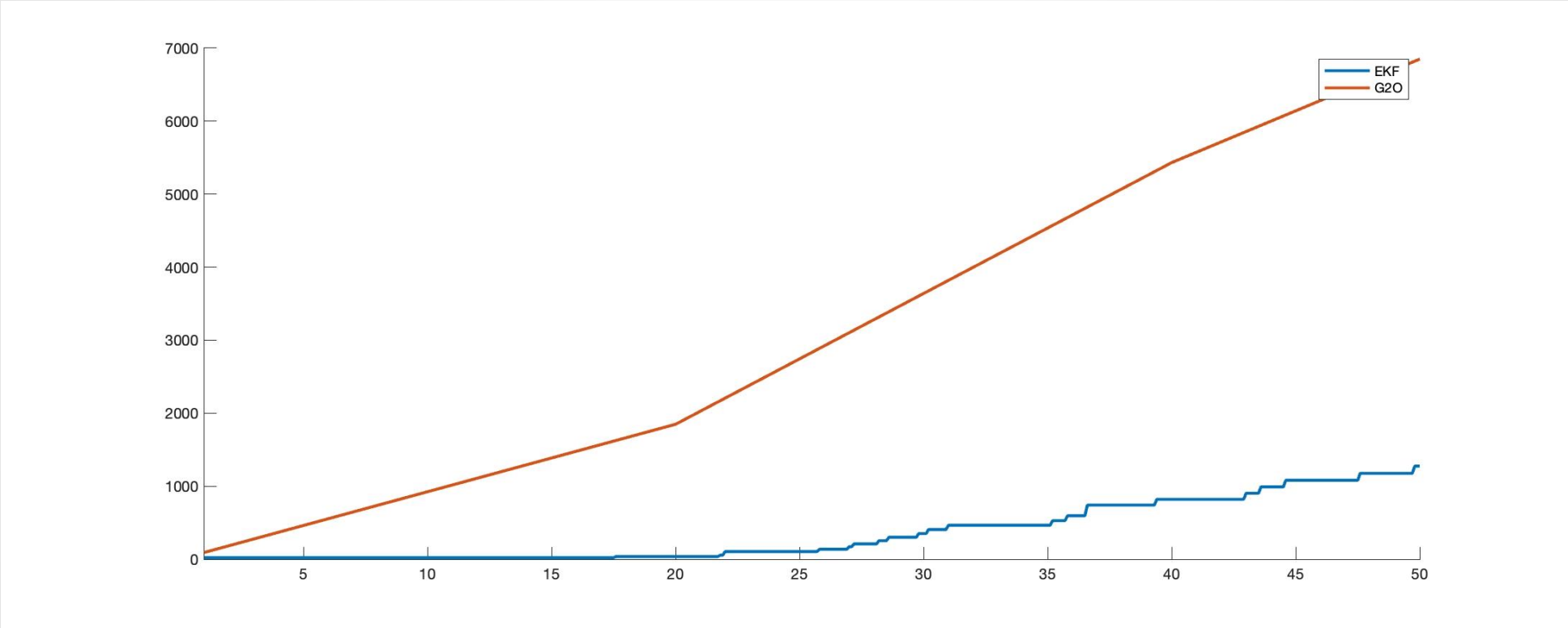
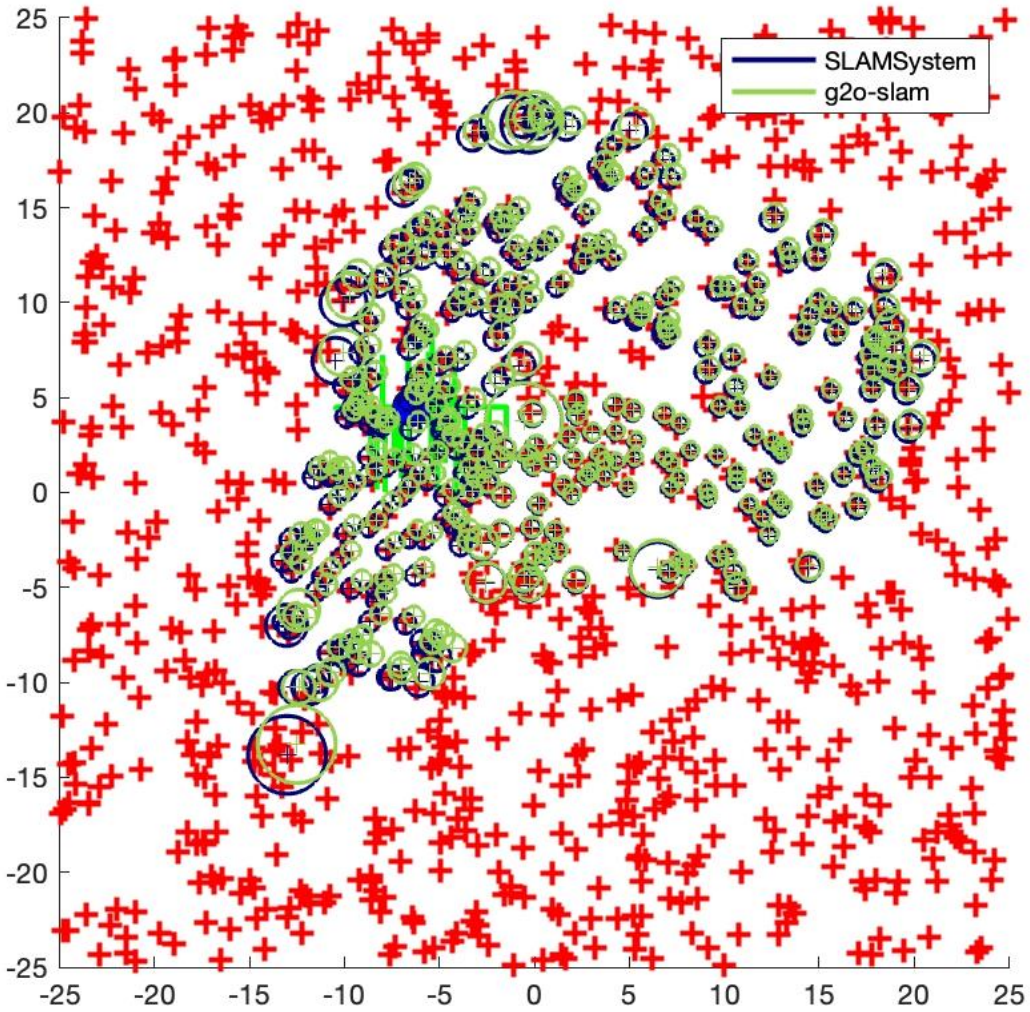
Small Scenario Storage




SLAMSystem
g2o-slam
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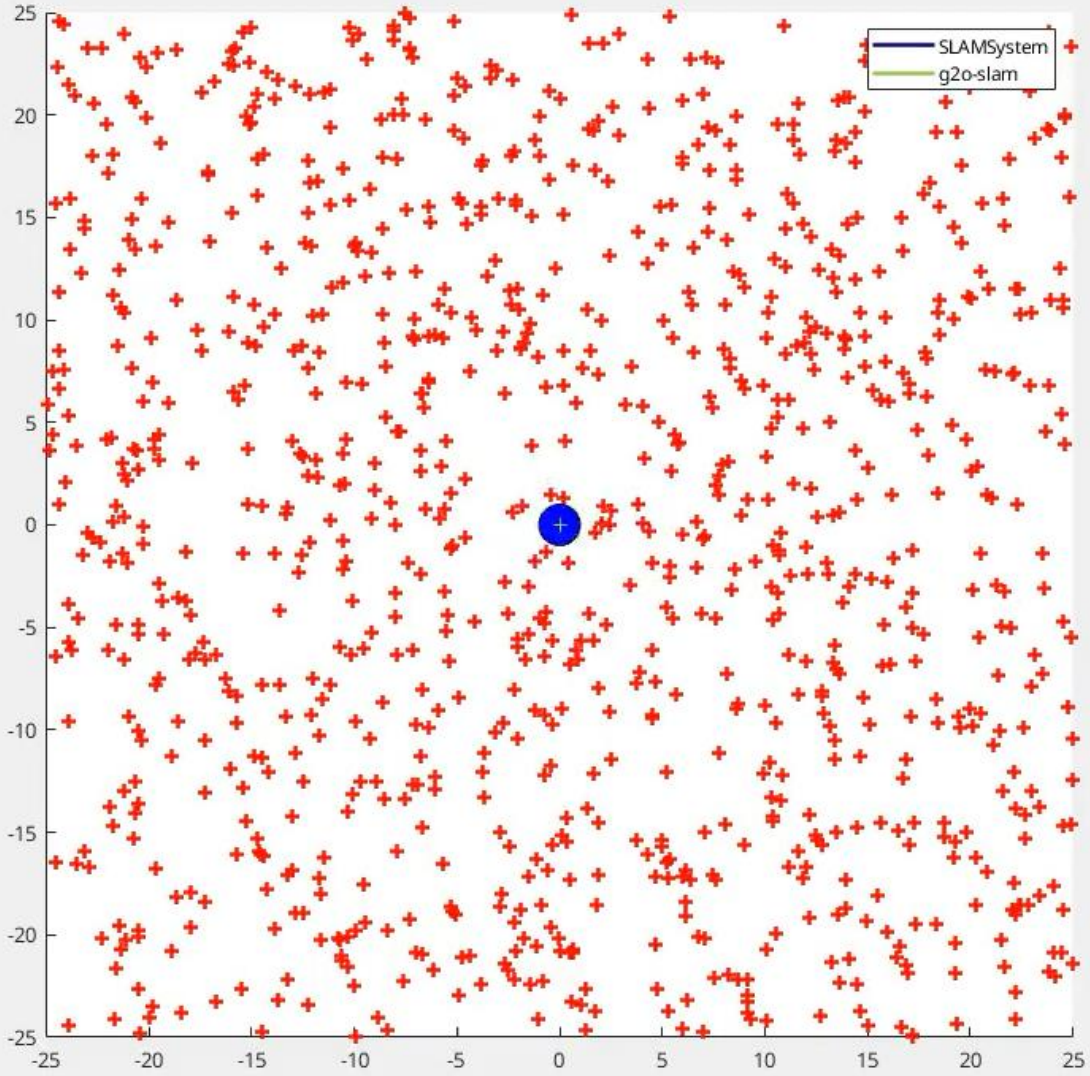
SLAMSystem
g2o-slam
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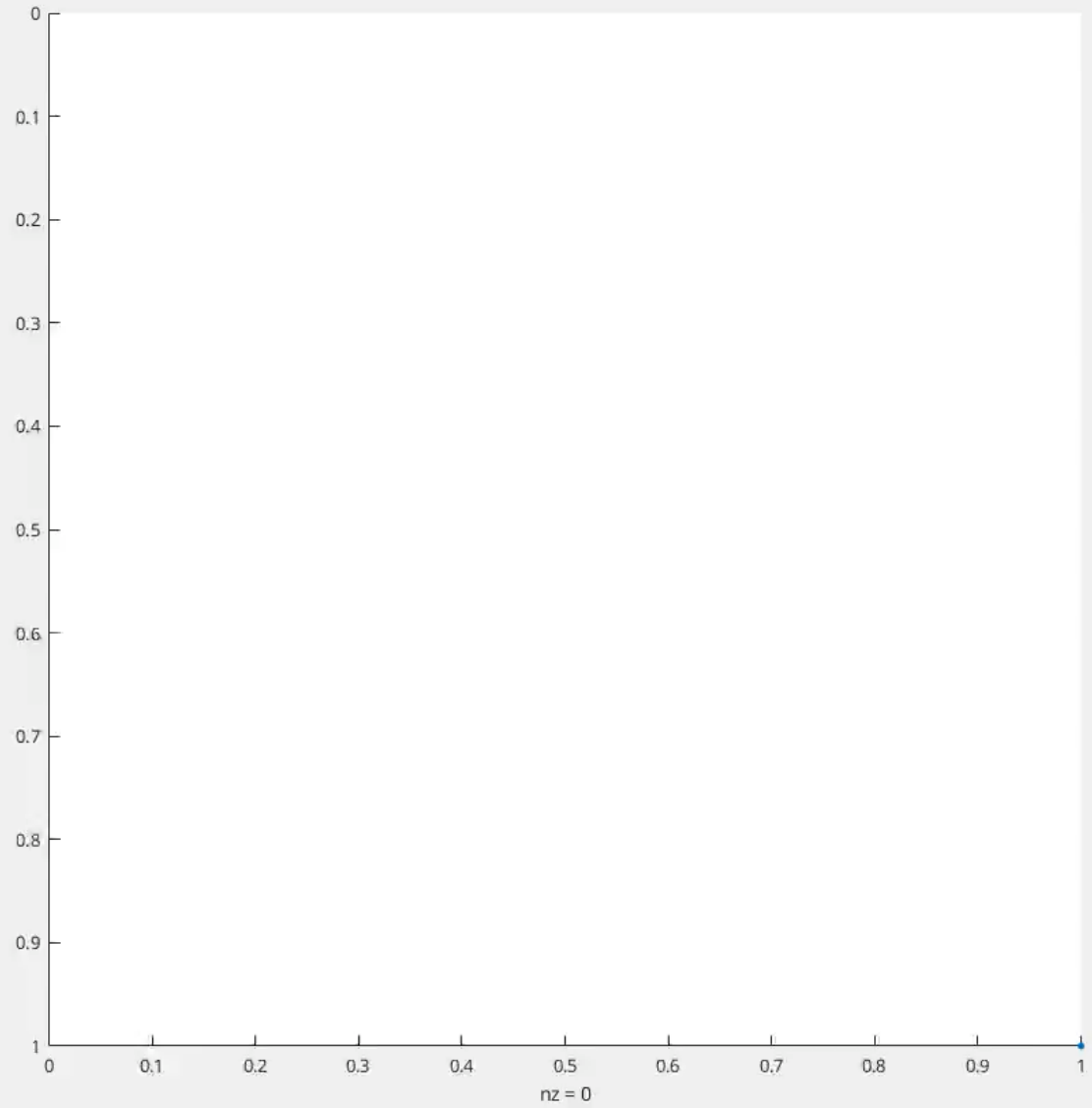
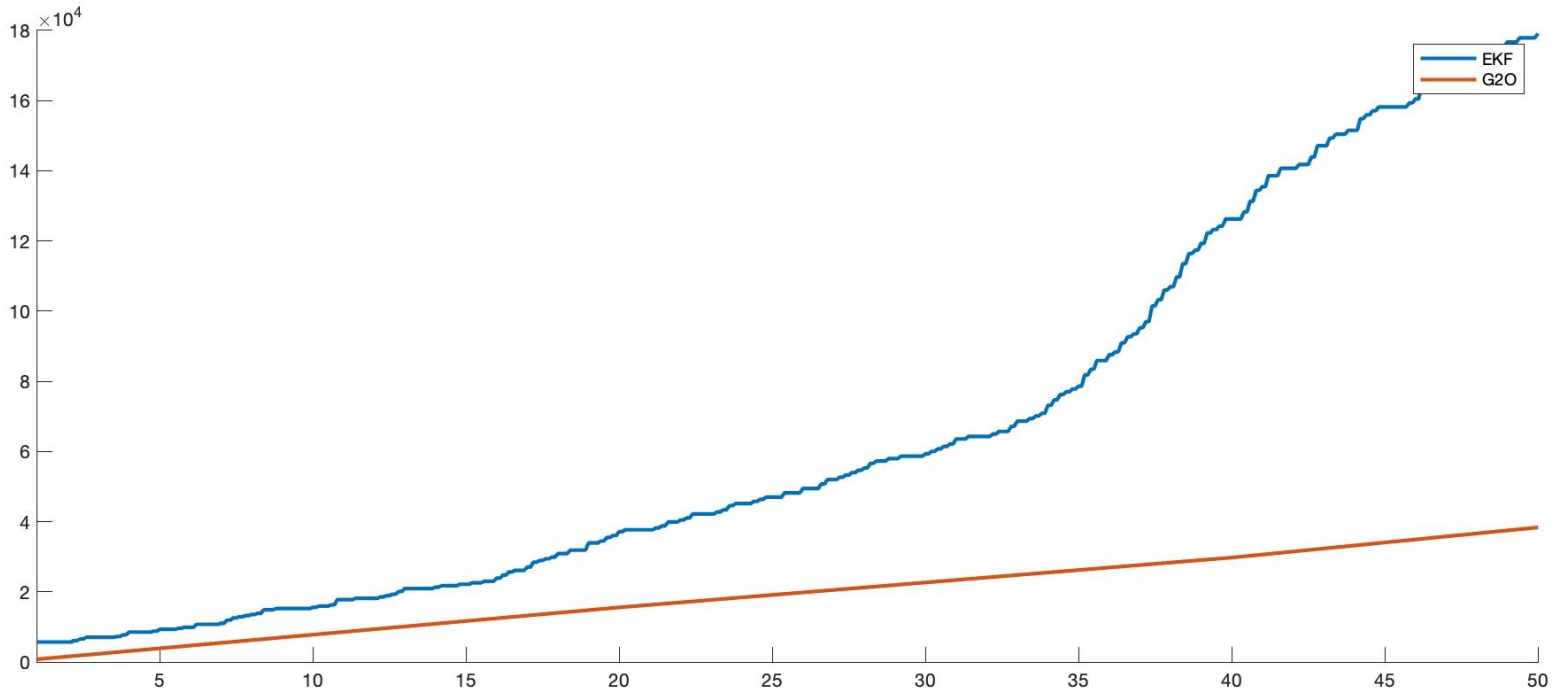

0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
nz = 0

Large Scenario Storage



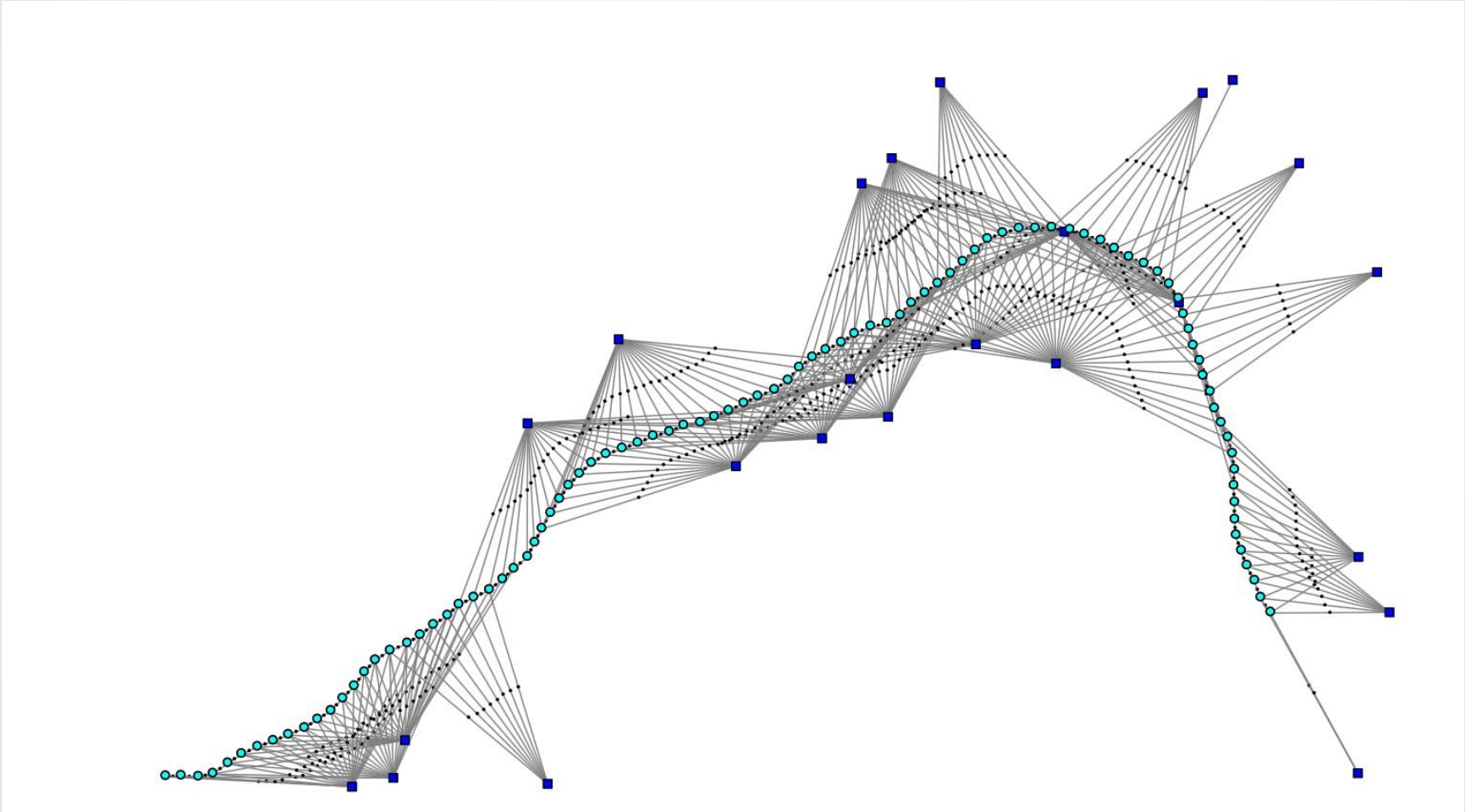
Improving Computation Further

- There are a lot of strategies we can use:
  - Selective optimization
  - Graph pruning
  - Controlling how landmarks enter the map

Selective Optimization

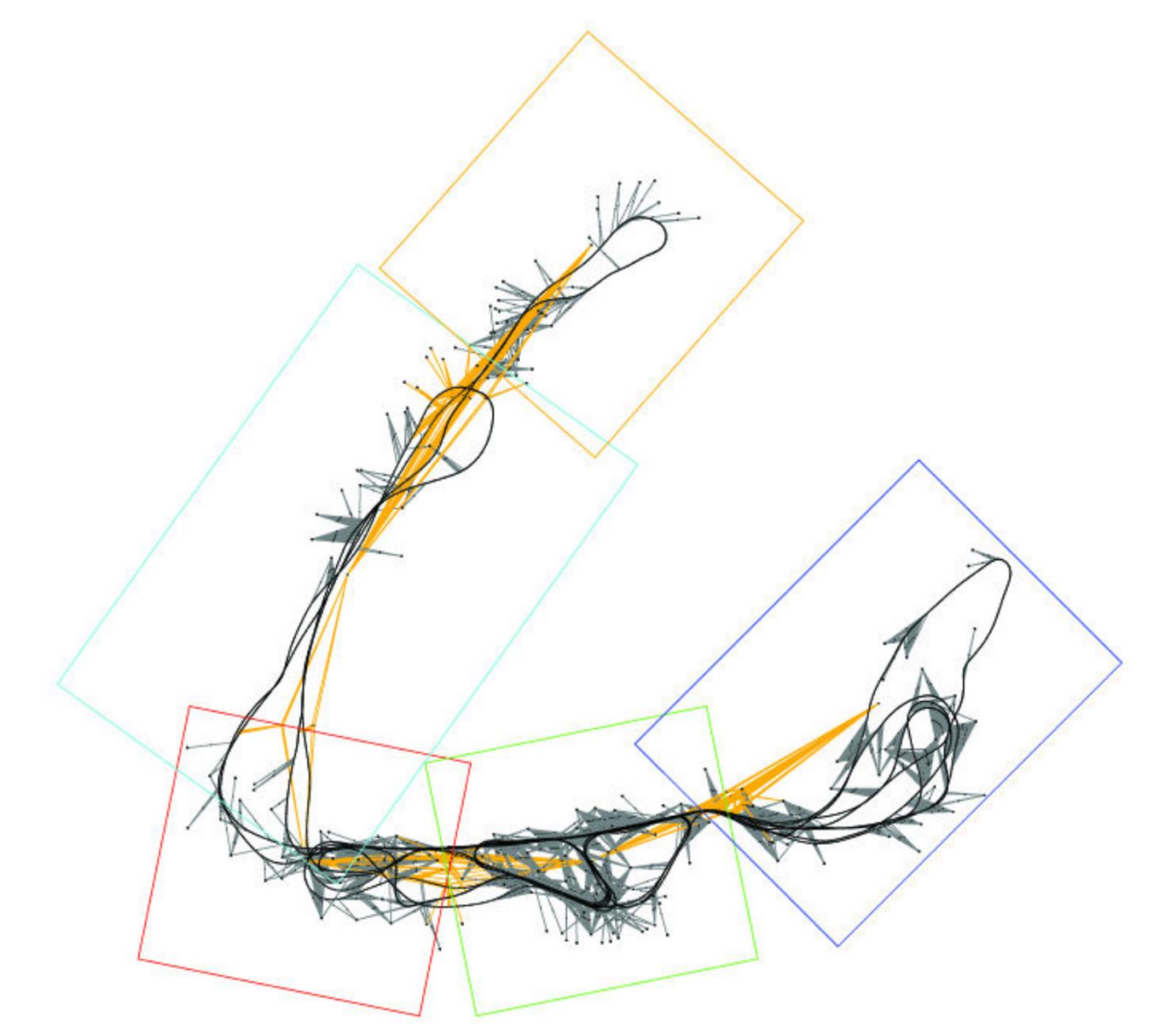
- You don’t have to optimize the whole graph at every time step
- You can choose to update just a subset of the vertices, which greatly reduces the costs

Fixing Vertices



Selective Optimization

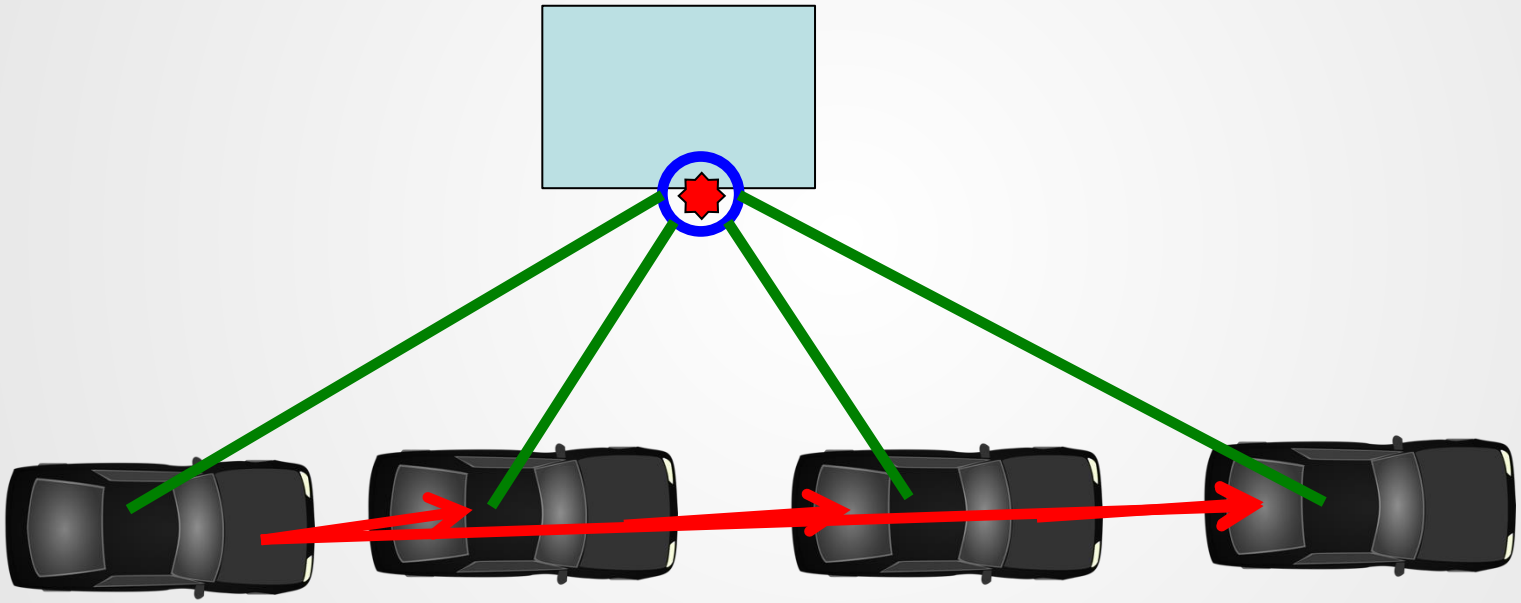
- You don’t have to optimize the whole graph
- Rather, you can choose to update just a subset of the vertices, which greatly reduces the costs
- There are various schemes, but mathematically they are the same as *fixing* a vertex
- When a vertex is fixed, its value is held constant and you can avoid ever putting it into the optimization step



Graph Pruning

- Any landmark will be observed multiple times by a vehicle
- We've already seen that the covariance on the landmarks falls more slowly the more you observe them
- Therefore, you can end up with a lot of observations (=platform states, edges) which contribute little to improving the overall accuracy

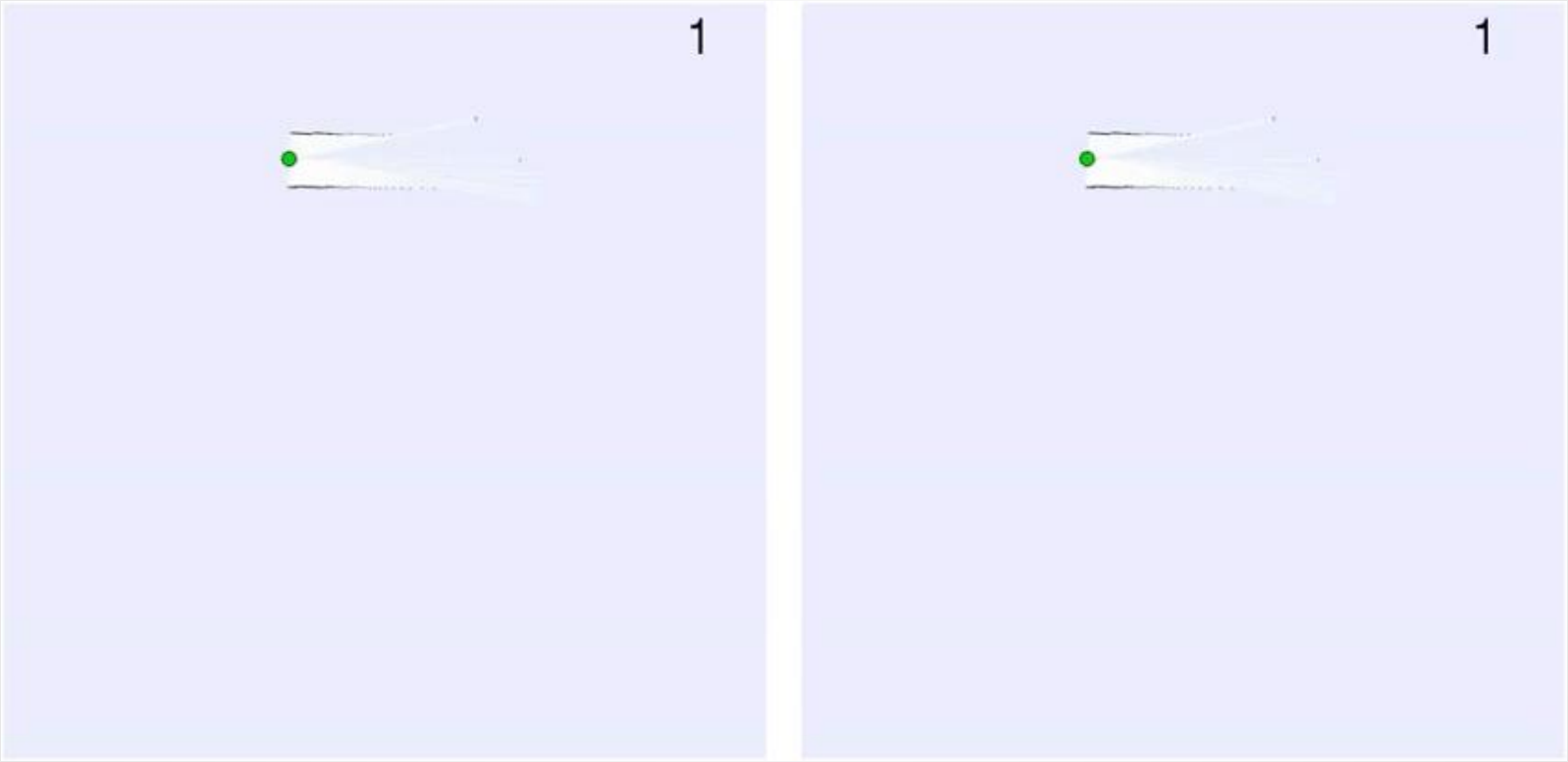
Graph Pruning



Graph Pruning through Marginalization

- One way to delete a vertex or an edge is to simply remove it from the graph
- However, this causes a loss of information
- Therefore, a lot of approaches marginalize out vertices instead, often using various Gaussian assumptions

Pruning Vertices from the Map

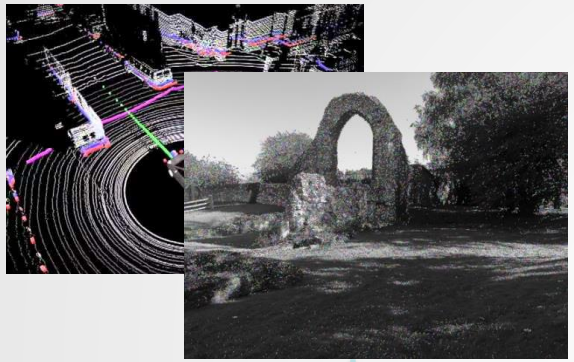


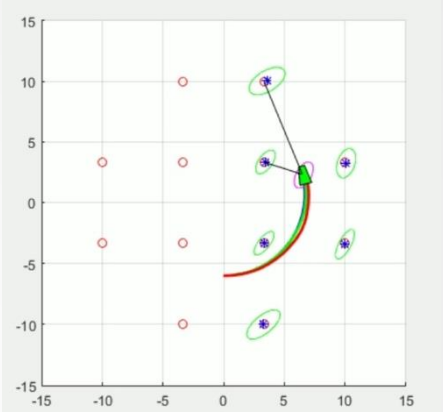
From “SLAM - Graph-based SLAM with Node Reduction”

Restricting What Goes into the Map

- Another approach is to restrict what get puts into a map in the first place
- A very common approach is to use a gateway and have an idea of key-frame based SLAM
- A frontend and a back end are used

Two End Design Limits What Goes In





Frontend

Feature Map

Backend

Subset of "interesting" information

Summary

- The application of factor graphs to SLAM is very easy:
  - You create new vertices for each new landmark
  - You connect edges whenever a landmark is observed
- These store the full dependency structure, but in a much simpler way (for the end user!)
- This is all very theoretical though
- Let's start seeing how to build vision-based SLAM systems in practice